03 Ch 03 — Advanced Capital Budgeting Decisions

Asked in last 3 papers: Sep 2025 MCQ CS-I (6m) sensitivity — % change in NPV for a 10% adverse change in cost, inflow, cost of capital · Jan 2026 MCQ CS-V (6m) expected value of profit from a probability distribution · May 2026 — nil.

1. Current trends in capital budgeting

→ Intermediate level assumed **no risk, cash flows certain**. In practice projects are exposed to: **(i) inflation, (ii) change in technology, (iii) change in Government policies**.

1.1 Impact of inflation on capital budgeting decisions

Inflation raises revenues and costs. Two distortions:

→ **(1) Depreciation is on historical cost** → tax shield on depreciation does not keep parity with inflation. Effect: inflation **reduces the actual (real) rate of return**.

→ **(2) Cost of capital contains a premium for anticipated inflation** → if k is nominal, cash flows must also be inflated; if no inflation, discount at the real rate.

Formula	Terms / logic	SM example figures
$CFAT = (R - C - D)(1 - T) + D =$ **$(R - C)(1 - T) + DT$**	R revenue, C cash costs (excl. depreciation), D depreciation, T tax rate. $(R - C)(1 - T)$ moves with inflation; **DT does not** (historical cost).	Example: outlay 80,000, 4 yrs, R 60,000, C 20,000, T 50%, D 20,000 → CFAT 30,000 p.a. With 10% inflation: 32,000 / 34,200 / 36,620 / 39,282; deflated (real) 29,091 / 28,264 / 27,513 / 26,830 — **less than 30,000**.
$R_N = R_R + P$	R_N nominal required return; R_R real required return; P anticipated inflation. Investors want real return plus inflation.	—
Nominal Cash Flow = (1 + Inflation Rate) Real Cash Flows	Apply once per year, i.e. (1 + Inflation Rate)^t for year t. Discount nominal cash flows at the nominal rate.	Illus 1: 15,000 × 1.10 = 16,500; × 1.10^2 = 18,150; 19,965; 21,962.
Real Discount Rate = (1 + Nominal Discount Rate) ÷ (1 + Inflation Rate) − 1	Discount real CF at real rate. Both routes give the same NPV — never mix real CF with nominal k.	Illus 1: 1.12 ÷ 1.10 − 1 = 0.0182 i.e. 1.8%.
$N.P.V. = {}^{n}\Sigma_{t=1}$ $[\{R_t(1 + i_r) - C_t$ $^{t}\Sigma_{r=1}(1 + i_c)\}(1 - T) + D_tT] \div (1 + k)^t - I_0$	R_t, C_t revenues / costs for year t with no inflation; i_r, i_c annual inflation rate of revenues / costs for r^{th} year; T tax rate; D_t depreciation; I_0 initial outlay; k cost of capital (with inflation premium). In working, the yearly rates are compounded cumulatively (Illus 2: × 1.10 × 1.09 ...); DT is not inflated.	Illus 2 (below).

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Illus 1: outlay 40,000, real CF 15,000 × 4 yrs, k 12% incl. 10% inflation. (i) Nominal CF at 12%: 14,732 + 14,469 + 14,211 + 13,957 − 40,000 = **₹ 17,369**. (ii) Real CF at 1.8%: 57,367 − 40,000 = **₹ 17,367** (approx.).

Illus 2 (XYZ Ltd): outlay 8,00,000, D 2,00,000, T 60%, k 10%. Inflated revenue 6,60,000 / 8,39,300 / 10,35,936 / 11,08,452 (e.g. 7,00,000 × 1.10 × 1.09); costs 3,36,000 / 4,92,800 / 5,37,152 / 5,80,124. Tax shield 2,00,000 × 0.60 = 1,20,000. Cash inflow 2,49,600 / 2,58,600 / 3,19,514 / 3,31,331; PV 9,06,744 → NPV **₹ 1,06,744**.

TYK P17: nominal CF (70), 30, 40, 30 lakhs; k 10% (taken as real), inflation 5%. Real CF = nominal × $1/1.05^t$: 28.56, 36.28, 25.92 → PV 75.40 − 70 = NPV **+5.40**. Alt: nominal rate 15.50% (PVF 0.866, 0.749, 0.649) → NPV 5.41. Viable.

TYK P18 (Shashi): same flows, outlay 72. Adjusted PVF 0.909/1.05 = 0.866; 0.826/1.1025 = 0.749; 0.751/1.1576 = 0.649 → PV 75.41 − 72 = NPV **3.41 lakhs**. Alt: nominal k = (1.10)(1.05) − 1 = 15.50% → NPV +3.45 (PVF 0.750 used for year 2).

TYK P19 (KLM): outlay 15,00,000, D 5,00,000, T 35%, nominal k 14%. Revenue 10,90,000 / 15,30,360 / 17,46,965; costs 5,50,000 / 7,19,400 / 8,33,905; PAT 3,51,000 / 5,27,124 / 5,93,489; + shield 1,75,000 → inflows 5,26,000 / 7,02,124 / 7,68,489; PV 15,19,965 → NPV **₹ 19,965**.

TYK T3 — Project appraisal under inflation: costs rise on all heads (labour, material, fixed assets, remuneration); purchasing power and demand fall; lenders revise rates. Guidelines (used by Govt agencies, banks, FIs):

- → (i) Provide for **cost escalation** on all heads, for the likely period of delay in implementation.
- → (ii) Scrutinise **sources of finance** for probable revision of interest rates → cost of funds goes up.
- → (iii) Adjust **profitability and cash-flow projections** for inflationary pressures.
- → (iv) Examine **financial viability at revised rates**; economic rate of return should accommodate the inflation rate.
- → (v) Prefer projects with **early payback** — long payback projects are riskier.
- → Two approaches: (a) adjust each year's cash flows to an **inflation index** (price and cost increases); or (b) adjust the **acceptance (cut-off) rate**, keeping cash flows at current prices. ICAI example: normal 15%, inflation 5% → "15.0 × 1.05 or 15.75%". Measuring inflation has no standard approach.

Note (app-written): ICAI's TYK T3 example multiplies the rate (15 × 1.05 = 15.75%). On the SM's own Illus 1 logic the inflation-adjusted rate would be (1.15)(1.05) − 1 = 20.75% (or 15 + 5 = 20% by $R_N = R_R + P$). Keep ICAI's text for theory; for numericals use the Illus 1 / TYK P18 method.

1.2 Impact of change in technology

Why analyse (qualitative factor that affects future cash flows):

- → Can significantly **alter the production process**.
- → Benefits: improved **quality, delivery time, flexibility**.
- → Can **reduce cost of capital**; can **improve cash inflows**.
- → May need **additional capital expenditure**.
- → Sales volume hit as **product life cycle shortens** (consumer preference).

How to incorporate (review the project continuously):

- → 1. At decision time, consider technology risk by **sensitivity, scenario, simulation** analysis.
- → 2. After launch, analyse positive/negative impact and **revise estimates** in money terms.
- → 3. If unviable, evaluate the **abandonment option**.
- → 4. Suitably **adjust the discount rate**.

1.3 Impact of change in Government policies

→ External factor; affects cash flows directly or through suppliers, buyers, customers, service providers. Impact can be +ve or −ve → revise estimates; if needed consider **abandonment**.

→ Some policies change over 5–10 years (e.g. **New Industrial Policy 1991**); others within a year:

→ **Fiscal policy** — government spending, taxation and borrowing to influence economic activity, aggregate demand, output, employment.

→ **Monetary policy** — central bank instruments to regulate availability, cost and use of money and credit (open market operations, reserve requirements, selective credit controls, currency supply, bank/repo rates). MPC meets at least **four times a year** → changes 2–3 times a year.

1.3.1 Domestic capital budgeting

→ (a) **Interest rates** (monetary policy) → cost of debt / YTM → cost of capital. Long-term projects less affected, but **working capital** (bank overdraft) is; small change ignored, a major change (about **100 basis points**) calls for revising cash flows.

→ (b) **Fiscal policy** → tax rate and depreciation rate → annual cash flows; drastic change → revise estimates.

1.3.2 International capital budgeting

→ (a) Bank rate and money supply changes move the **foreign exchange rate** → revise estimates.

→ (b) Change in **tax on foreign income** or **DTAA** provisions (fiscal policy) → revise estimates.

→ Conclusion: forecast policy changes and build a provision into expected cash flows.

2. Dealing with risk in investment decisions

→ (i) **Certainty** — cash flows certain.

→ (ii) **Risk** — cash flows risky and **probability can be assigned**.

→ (iii) **Uncertainty** — probability **cannot** be assigned.

2.1 Risk and uncertainty

→ **Risk** = variability of actual returns vs estimated returns; measured mostly by **SD and coefficient of variation**. Risk: probability distribution known. Uncertainty: no information to form it. Terms used interchangeably.

2.2 Reasons for adjusting risk

→ 1. **Opportunity cost** for the level of risk — are returns proportionate to risk, and better than other options?

→ 2. To know the **real value of cash inflows** — higher risk → higher risk premium → higher expected return.

3. Internal and external factors affecting capital budgeting decisions

Factor	What (SM)	SM example
3.1 Internal		
3.1.1 Project-specific	Completion in time, errors in estimating resources and allocation, cash flow estimation.	Nuclear vs hydel power project.
3.1.2 Company-specific	Credit rating downgrade, change in key managerial persons, IPR/law violation cases, disputes with workers — affect cash flows and access to funds.	Two banks, different default risk.
3.2 External		
3.2.1 Industry-specific	Regulatory restrictions on industry, changes in technology.	Leather, breweries.
3.2.2 Market risk	Entry of substitutes, demand changes, availability and access to resources.	Thermal plant without coal supply.
3.2.3 Competition	Entry of rival, product dynamism, change in tastes and preferences.	—
3.2.4 Economic conditions	Monetary and fiscal policy (new taxes, cess), inflation, GDP, savings, disposable income.	—
3.2.5 International	Restrictions on free trade / market access, recessions, bilateral agreements, political and geographical conditions.	Restriction on outsourcing jobs overseas.

4. Methods of incorporating risk in capital budgeting

Statistical techniques (4.1)	Conventional techniques (4.2)	Other techniques (4.3)
Probability; Variance / SD; Coefficient of variation	Risk-adjusted discount rate; Certainty equivalent	Sensitivity; Scenario; Simulation; Decision tree

4.1 Statistical techniques — 4.1.1 Probability

Asked: Jan 2026 MCQ CS-V 13-15 6m

→ Probability = chance an event occurs; certain event = **1**, impossible = **0**; probabilities of all outcomes add to 1.

Formula	Terms / logic	SM example figures
$E(R)/ENCF = \Sigma_{i=1..n} NCF_i \times P_i$	NCF_i net cash flows, P_i probability. Expected (mean) cash flow: each possible cash flow × its probability, added.	Example: 3,00,000 × 0.3 + 2,00,000 × 0.6 + 1,20,000 × 0.1 = 90,000 + 1,20,000 + 12,000 = **2,22,000**.
$ENPV = \Sigma_{t=1..n} ENCF \div (1 + k)^t$	t period, k discount rate. Find ENCF year by year, then discount at k. ENCF includes inflows and outflows.	Illus 3, 4 below.

Note (app-written): the SM text after the first table says chances of "₹ 1,00,000" are 10%; the table and the working use ₹ 1,20,000 (SM p.13).

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Illus 3 (single period): outlay 10,000, k 10%. ENCF A 12,000, B 16,000. NPV A = 0.909 × 12,000 − 10,000 = **₹ 908**; B = 0.909 × 16,000 − 10,000 = **₹ 4,544** → B.

Illus 4 (multi-period): ENCF yr 1 6,000, yr 2 4,800, yr 3 4,200; PV = 6,000 × 0.909 + 4,800 × 0.826 + 4,200 × 0.751 = 12,573 − 10,000 = ENPV **₹ 2,573**.

TYK P21 (Jumble — expected utility): same idea with utilities: Σ utility × probability. A = −10 − 12 + 16 + 6 + 2 = **2**; B = −6 − 0.45 + 16 + 5 + 3 = **17.55** → select **B** (higher expected utility).

4.1.2 Variance · 4.1.3 Standard deviation · 4.1.4 Coefficient of variation

Formula	Terms / logic	SM example figures
Variance $(\sigma^2) = \Sigma_{j=1..n} (NCF_j - ENCF)^2 P_j$	Variance: squared deviation of each outcome from the mean, weighted by probability. Zero = cash flows fixed (e.g. fixed-sum service contract); large = innovative project; small = established market.	Illus 5: A 48,00,000; B 1,76,00,000.
Standard Deviation (σ) = √Variance (σ^2)	SD: risk in the estimated cash flows. Variance is a prerequisite for SD.	Illus 5: A **2,190.90**; B **4,195.23**.
Coefficient of variation = Standard Deviation ÷ Expected Return / Expected Cash Flow	Risk per unit (per rupee) of expected return. Use when projects have **different expected values**. Lower CV = better risk-return → select.	Illus 6: A 2,190.90 ÷ 14,000 = **0.1565**; B 4,195.23 ÷ 18,000 = **0.2331** → A.
Hillier's model: $M = \Sigma_{i=0..n} (1 + r)^{-i} M_i$; $\sigma^2 = \Sigma_{i=0..n} (1 + r)^{-2i} \sigma_i^2$	M mean of PV, M_i expected cash flow of year i, σ_i SD of year i, r discount rate. Multi-year, **independent** cash flows: discount each year's variance by $(1 + r)^{2i}$, add, take root. Used in ICAI TYK PI, PS.	TYK PI: $\sqrt{(85.4/1.06^2 + 98.61/1.06^4 + 74.29/1.06^6)}$ = 14.3696.

→ **Importance:** both measure volatility of estimated cash flows; **variance** = range of variability; **SD** = risk of the proposal. Cash flows close to the mean → less risky.

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Illus 5: ENCF A 12,000, B 16,000; σ^2 A = 16,00,000 + 8,00,000 + 0 + 8,00,000 + 16,00,000 = 48,00,000 → σ 2,190.90; B 1,76,00,000 → σ 4,195.23. **Illus 6:** ENCF A 14,000, B 18,000; same σ; CV A 0.1565 (risk ₹ 0.16 per rupee), B 0.2331 (₹ 0.23) → **A better**.

TYK P1 (Skylark, ₹ lakhs): outlay 50, risk-free rate 6%. ENCF 27.0 / 29.3 / 27.9 → PV 25.461 + 26.077 + 23.436 = 74.974 → ENPV **24.974**. σ_1 = √85.4 = 9.241; σ_2 = √98.61 = 9.930; σ_3 = √74.29 = 8.619; σ of PV = **14.3696**. (iii) SD measures dispersion around the mean; identifies least risky of projects with similar mean; lower CV preferred if sizes differ; if distribution is approx. normal → probability of NPV above/below a value.

TYK P2 (Cyber): ENCF both 4,00,000; PVAF 7%, 5 yrs 4.100 → NPV both **11,40,000**. σ A 1,54,919.33, B 77,459.66 → **B** (same NPV, lower SD).

TYK P3: SD → X (90,000 < 1,20,000). CV X = 90,000 ÷ 1,22,000 = **0.738**; Y = 1,20,000 ÷ 2,25,000 = **0.533** → Y. (iii) In conflict, **NPV method is best** — consistent with wealth maximisation.

TYK P4 (KLM): K: ENCF 15, σ^2 4.8, σ 2.19, CV **0.146**. S: ENCF 17, σ^2 20.8, σ 4.56, CV **0.268** → S riskier.

TYK P5 (XY Co, ₹ lakhs, 10%): X: ENCF 48.5 / 41.5 / 38.5 → PV 107.28 → NPV **37.28**. Y: 45.5 × 2.487 = 113.16 → NPV **33.16** → X on NPV. Hillier: X σ_i^2 185.25, 95.25, 35.25 → σ = √238.06 = **15.43**; Y σ_i^2 12.25 each → √25.4 = **5.03** → Y less risky.

TYK P6 (Shivam): ENPV A = B = **9,000**. σ A √1,98,00,000 = ₹ 4,450; B √1,44,00,000 = ₹ 3,795. PI = (ENPV + cost) ÷ cost: A 45,000/36,000 = **1.25**; B 39,000/30,000 = **1.30** → **B** (same EV, less dispersion).

Note (app-written): TYK P6 (ii) labels Project B's SD as "Standard Deviation of Project A = ₹ 3,795" — it is Project B's (SM p.64).

4.2 Conventional techniques — 4.2.1 Risk-adjusted discount rate (RADR)

→ Investors demand **higher returns from riskier projects**. Required return = compensation for delaying consumption + inflation (= risk-free rate) + compensation for risk. Riskier than a similar project → adjust discount rate **upward**.

→ **Risk-free rate**: return on investments bearing no risk (e.g. Govt securities 6%). **Risk premium**: return over risk-free rate for bearing extra risk; depends on investor's risk perception and risk aversion.

Formula	Terms / logic	SM example figures
RADR = Risk-free rate + Risk premium	k used in the numerator-unchanged NPV.	Illus 7: 7% + 7% = 14%.
$NPV = \Sigma\ NCF_t \div (1 + k)^t - I$	NCF_t net cash flow; k = RADR; I initial investment.	Illus 7: NPV at 7% 144.34; at 14% 99.79 lakhs.
Required Rate of Return = $IR_f + (k_o - IR_f)$ Risk Factor	ICAI TYK P16 (on the basis of CAPM): IR_f risk-free rate, k_o firm's minimum required rate of return, Risk Factor = project's risk index.	TYK P16: 10% + (15% − 10%) × 1.80 = 19%.

→ **Advantages:** (1) **easy to understand**; (2) incorporates **risk premium in the discounting factor**.

→ **Limitations:** (1) difficulty in finding **risk premium and RADR**; (2) NPV can be found but **not the SD** of the project.

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Illus 7: outlay 100 lakhs; CF 25, 60, 75, 80, 65. At risk-free rate 7%: PV 244.34 → NPV **144.34**. At RADR 14%: PV 199.79 → NPV **99.79 lakhs**.

TYK P15: RADR by CV (1.2 → 16%, 0.8 → 14%, 0.4 → 12%). X: 70,000 × 3.274 = 2,29,180 − 2,10,000 = **19,180**; Y: 42,000 × 3.433 = 1,44,186 − 1,20,000 = **24,186**; Z: 30,000 × 3.605 = 1,08,150 − 1,00,000 = **8,150**.

TYK P16 (New Projects): RADR P-I 19%, P-II 15%, P-III 13%. NPV P-I 6,00,000 × 2.639 − 15,00,000 = **83,400**; P-II 12,67,800 − 11,00,000 = **1,67,800**; P-III 21,13,800 − 19,00,000 = **2,13,800** → **P-III**.

Note (app-written): TYK P16 gives a 40% tax rate; ICAI's answer does not use it (inflows discounted as given).

4.2.2 Certainty equivalent (CE)

- **CIMA:** express risky future cash flows as the **certain cash flow** the decision maker considers equivalent — indifferent between the risky amount and the (lower) riskless amount. Guaranteed return management would accept instead of a higher uncertain one; incorporates the decision maker's **utility function**.
- Coin example: win 10,000 on head, nil on tail → EV 5,000; indifferent at certain 3,000 → CE 3,000, coefficient 3,000 ÷ 10,000 = **0.3**.

Formula	Terms / logic	SM example figures
α_t = Certain cash flow ÷ Risky or expected cash flow	CE coefficient lies **between 0 and 1**; 1 = certain cash flow or management risk-neutral. Higher risk → lower α.	Illus 8: 0.90, 0.85, 0.82, 0.78.
$NPV = \Sigma\ \alpha_t \times NCF_t \div (1 + k)^t - I$	α_t risk-adjustment factor (CE coefficient); NCF_t forecast net cash flow for year t without risk-adjustment; **k = risk-free rate** (constant); I initial investment. Risk already in numerator → using cost of capital = **double counting**.	Illus 8: NPV ₹ 5,34,570.

- **Steps:** 1. Multiply each risky CF by α_t. 2. Discount at **risk-less rate**. 3. Apply normal method; under IRR, compare IRR with the **risk-free rate**, not the required return.
- **Advantages:** 1. **simple**, easy to understand and apply; 2. easily applied to **different risk levels for different cash flows** (a riskier year can be adjusted and NPV recalculated).
- **Disadvantages:** 1. **no objective/mathematical method** — CEs subjective; 2. decided on **management's** risk perception; shareholders' risk perception ignored → not used often.
- **RADR vs CE:** CE is **superior** — does not assume risk increases with time at a constant rate; each year's α reflects that year's risk. But RADR is **preferred in practice** — difficult to specify a series of α's, simple to adjust the discount rate.
- **TYK T1 (short note):** adjusts estimated cash flows, discounts at risk-free rate; converts expected flows to riskless amounts; greater the risk, smaller the CE for receipts and larger for payments; superior to RADR as it measures risk more accurately; adjusts cash flows not discount rates; explicitly recognises risk but the reduction procedure is implicit and may be inconsistent between investments.

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Illus 8: cost 45,00,000, risk-free rate 5%: $10,00,000 \times 0.90/1.05 + 15,00,000 \times 0.85/1.05^2 + 20,00,000 \times 0.82/1.05^3 + 25,00,000 \times 0.78/1.05^4 - 45,00,000$ = **₹ 5,34,570**.

TYK P14 (Textile): risk-free rate 6%. M: CE flows 3,60,000 / 3,50,000 / 2,50,000 → PV 8,60,980 − 8,50,000 = **10,980**. N: 4,05,000 / 3,60,000 / 3,50,000 → PV 9,96,315 − 8,25,000 = **1,71,315** → accept **N**. (ii) Sum of CE coefficients M 2.0 < N 2.4 → M riskier → under RADR, **M at a higher rate** (positive correlation between risk and discount rate).

4.3 Other techniques — 4.3.1 Sensitivity analysis

Asked: Sep 2025 MCQ CS-1 1-3 6m

→ **CIMA:** modelling and risk-assessment procedure in which changes are made to significant variables to see the effect on the planned outcome; attention then on variables of special significance.

→ Change **only one variable at a time** (others constant). The more sensitive the NPV (or IRR), the **more critical** the variable.

→ **Steps:** 1. find variables that influence NPV/IRR; 2. establish mathematical relationship among them; 3. analyse effect of change in each on NPV/IRR.

Formula	Terms / logic	SM example figures
Change in NPV = (NPV − NPV (Revised)) ÷ NPV	As ICAI TYK P10 working. Method 1: change one variable by a given % adversely (any %, e.g. 10%), recompute NPV. Highest % change = most sensitive.	TYK P10: inflow −10% → (22,605 − 8,345) ÷ 22,605 = 63.08%.
% change in a variable to make NPV = 0 = NPV ÷ PV of that variable's cash flows × 100	App-written summary of ICAI's TYK P9/P11 working. Method 2 (break-even): how far the variable can move before NPV becomes zero. **Smallest %** = most sensitive.	TYK P11: savings 4,914 ÷ 22,792 = 21.56%; plant 4,914 ÷ 10,000 = 49.14%.
(IRR − X) ÷ X = cost of capital sensitivity	X = cost of capital; % rise in X before NPV = 0. ICAI TYK P12 writes it with figures; general form app-written.	(16 − X) ÷ X = 60% → X = 10%.

→ **Advantages:** (1) **Critical issues** — identifies critical factors for success/failure; (2) **Simplicity**.

→ **Disadvantages:** (1) **Assumption of independence** of variables — unlikely in real life; (2) **ignores probability** of changes.

→ **TYK T2:** measures how sensitive estimated parameters (cash flows, discount rate, life) are to estimation errors; studies responsiveness of NPV/IRR to selling price, quantity, returns; answers "what if" (e.g. flows ₹ 50,000 instead of ₹ 80,000; life 3 years not 5). Not a method to remove risk — only to analyse and measure it. Cash flows may be worst (pessimistic), most likely, most optimistic. Three steps: identify variables; define quantitative relationship; analyse impact on NPV.

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Illus 9 (X Ltd): *cost 400 cr, S cr units, SP 100, VC 50, FC 50 cr, 6%, 3 yrs (PVAF 2.673). Net inflow 200 cr → NPV* ***134.60****. 2.5% adverse: cost 410 → 124.60 (−7.43%); SP 97.5 → 101.19 (****−24.82%****); VC 51.25 → 117.89 (−12.41%); FC 51.25 → 131.26 (−2.48%); units 4.875 → 117.89 (−12.41%).* ***Selling price most sensitive.***

TYK P10: *cost 1,20,000, inflow 45,000, 4 yrs, 10% (3.169; 11% 3.103). NPV* ***22,605****. Cost +10% → 10,605 (****53.08%****); inflow −10% → 8,345 (****63.08%****); k 11% → 19,635 (****13.14%****) → most sensitive to* ***annual cash inflow****.*

TYK P9 (XYZ): *cost 10,00,000, margin 20, volumes 20,000/30,000/30,000, 10%. NPV 13,10,293 − 10,00,000 =* ***3,10,293****. Break-even: (a) SP: S − 40 = 10,00,000 ÷ 65,514 = 15.26 → S 55.26, fall* ***7.9%****; (b) unit cost 44.74, rise* ***11.85%****; (c) volume fall 3,10,293 ÷ 13,10,293 =* ***23.68%****; (d) outlay rise 3,10,293 ÷ 10,00,000 =* ***31.03%****; (e) life: 2 yrs NPV −1,40,800; yr 3 PV 4,50,600 ÷ 360 = 1,252 per day → 112 days → 2 years 112 days, fall* ***22.97%****. Alt (10% adverse): SP → NPV −82,796 (126.68%); cost → 48,234 (84.46%); volume → 1,79,263 (42.22%); outlay → 2,10,293 (32.22%).*

TYK P11 (Red Ltd): *NPV 14,914 − 10,000 =* ***4,914****. NPV-zero: plant* ***+49.14%****; running cost PV 7,878 →* ***+62.38%****; savings PV 22,792 →* ***−21.56%*** *→* ***savings most sensitive****. Alt 10%: 20.35%, 16.04%, 46.38%.*

TYK P12 (Easygoing): *(i) outlay = 3.274 × 57,500 =* ***1,88,255*** *(at IRR NPV = 0); (ii) k 10% → NPV 57,500 × 3.791 − 1,88,255 =* ***29,727.50****; (iii) 29,727.50 ÷ 3.791 = 7,841.60 = 7.8416% of FC → FC* ***1,00,000****; (iv) SP = 60 ÷ 30% = 200; sales = (57,500 + 1,00,000) ÷ 0.70 = 2,25,000 →* ***1,125 units****; (v) BEP = 1,00,000 ÷ 140 =* ***714.285 units****.*

TYK P13 (Unnat): *NPV = [2,00,000 × 13.50 − 10,00,000] × 3.605 − 50,00,000 =* ***11,28,500****. Break-even: SP 28.43 (****5.23%****); volume 1,76,812 (****11.59%****); VC 18.07 (****9.51%****). (d) Expected volume 1,95,000 → ENPV* ***8,85,163****; worst (1,75,000) −88,188; best 23,45,188. 30% chance of negative NPV > acceptable 20% →* ***reject****.*

Note (app-written): TYK P13(c) says variable cost 18.07 "represents a fall of 9.51%" — it is a rise from 16.50 (SM p.74).

4.3.2 Scenario analysis

- → *Extends sensitivity: brings in* ***probabilities*** *of changes in key variables and changes* ***more than one variable at a time****.*
- → *Start with* ***base case*** *(most likely); then* ***worst case*** *(low sales, low price, high VC) and* ***best case****; or mix some +ve and some −ve changes. Examines impact of alternative combinations of variables on NPV/IRR.*
- → ***Worst-case probability*** *(ICAI TYK P7, P8): cash flows* ***perfectly dependent*** *(positively correlated) over time → = probability of the low flow in year 1 (TYK P7: 0.3);* ***independent*** *→ that probability raised to the number of years (TYK P7:* $(0.3)^5$*) (general rule app-written from ICAI's figures).*

Sensitivity analysis	Scenario analysis
Impact of change in a ***single input*** *on NPV/IRR; others constant.*	*Based on a* ***scenario*** *(recession, boom, normal) —* ***all inputs change simultaneously****.*
Identifies the single ***critical variable*** *and range of outcomes.*	*Variability of outcome across scenarios shows project risk — higher deviation = higher risk.*
Simple.	***Far more complex****.*

ICAI ·

Illus 10 (XYZ, ₹ '000): outlay 1,400, 9%. NPV worst **−110.15**, most likely **100.85**, best **311.85**. Most likely yrs 1–2, worst yr 3: −14,00,000 + 5,04,587 + 3,78,756 + 5,40,528 = **₹ 23,871**.

TYK P7 (X Ltd): outlay 4,00,000, 10%. (i) Expected inflow 1,09,000 × 3.170 + 1,52,000 (incl. expected salvage 43,000) × 0.621 → ENPV **39,922**. (ii) Worst −8,580; best **92,060**. (iii) Worst-case probability: dependent **0.3**; independent $(0.3)^5$ = **0.00243**. (iv) Base case 47,950; ENPV 0.3(−8,580) + 0.5(47,950) + 0.2(92,060) = 39,813; σ **35,800**; CV = 35,800 ÷ 39,813 = **0.90**. (v) CV below 0.95–1.0 → less risky → k 10% − 1% = 9% → ENPV 3,53,160 + 98,800 − 4,00,000 = **51,960** → accept.

TYK P8 (XY Ltd): outlay 1,00,000, 20%, 5 yrs. Expected inflow 31,000 p.a., terminal 20,000 → NPV **₹ 746.52**. Worst (20,000, nil terminal) **−40,187.76**; best (40,000 + 30,000) **31,680.81**. Perfectly correlated → worst-case probability **10%**.

Note (app-written): TYK P7 shows ENPV 39,922 in (i) and 39,813 in (iv); the difference is ICAI's factors (3.170 + 0.621 for years 1–4 and 5 in (i) vs 3.790 for 1–5 in (ii)/(iv)) (SM p.65).

4.3.3 Simulation analysis (Monte Carlo)

- → Simulation = exact replica of the actual situation via a **model**; very many calculations give possible outcomes and their probabilities.
- → Monte Carlo ties together **sensitivities and probability distributions**; came out of the first nuclear bomb work; named after casino mathematics. Gives a **probability distribution of NPVs**, not a single point estimate.
- → Identify key factors and their inter-relationships; model cash receipts and payments; specify a range/probability distribution for each assumption.

4.3.3.1 Steps (TYK T4):

1. **Model the project** — NPV as a function of **parameters** (set by decision maker, constant over all runs) and **exogenous variables** (stochastic, outside decision maker's control).
2. Specify **values of parameters** and **probability distributions** of exogenous variables.
3. Select a value **at random** from each exogenous variable's distribution.
4. Determine **NPV** for the random values and pre-specified parameters.
5. **Repeat** steps 3 and 4 a large number of times.
6. Plot the **distribution of NPVs**; compute **mean and SD** to gauge risk.

- → **Random-number mapping:** cumulative probability → two-digit ranges (e.g. CF 10,000 p 0.02 → 00–01; 15,000 p 0.03 → 02–04; 20,000 → 05–19 …; life 3 yrs p 0.05 → 00–04; 4 yrs → 05–14; 5 yrs → 15–44 …).

ICAI ·

SM example: NPV = $\Sigma\ CF_t/(1+i)^t - I$; i 10%, I 1,30,000 are parameters; annual CF and life are exogenous. Ten runs; run 1: RN 53 → CF 30,000, RN 97 → 9 yrs, PVAF 5.759 → NPV **42,770**. Other runs: 85,075; (8,300); (66,600); (21,125); 40,380; 16,040; 16,040; 21,640; 650.

Note (app-written): the SM text says RN 53 gives "₹ 3,000"; the table maps 53 to ₹ 30,000 and the run uses 30,000 (SM p.32).

→ **4.3.3.2 Advantages** (strength lies in variability): (1) can **predict bad market situations** beforehand; (2) handles (a) **numerous exogenous variables** with any distribution, (b) **complex inter-relationships** among parameters, exogenous and endogenous variables that defy analytical methods, (c) **compels** the decision maker to consider inter-dependencies and uncertainties.

→ **4.3.3.3 Shortcomings:** (1) **difficult to model** and specify distributions; (2) **inherently imprecise** — rough approximation; misleading when a tail is critical; (3) complex model built by **management experts**, not the decision maker, who may not use it; (4) NPV in each run uses the **risk-free rate** (to avoid pre-judging risk) → NPV takes a different meaning, **hard to interpret**.

4.3.4 Decision tree analysis

→ For **sequential decisions** — present investment affects future decisions and events. Decision tree = **graphic display** of relationship between a present decision, future events, future decisions and consequences.

→ Two situations: **Decision** — manager has control; **Event** — no control → assign **probabilities** to outcomes. Manager acts rationally (most profitable in business; lowest cost in non-profit).

Steps:

1. **Define investment** — e.g. launch new product (local/national/international); extend factory or outsource; dig for oil and to what depth.
2. **Identify decision alternatives** clearly.
3. **Draw the tree** — cash flows, probabilities, expected PV on the branches. **Decision point = square** (action available); **event/chance point = circle** (with probabilities and values); drawn **left to right**.
4. **Evaluate** — from the **last decision (extreme right) back to the first (left)**: at each final decision point pick highest NPV, truncate others, assign that NPV to the point; compute expected NPV at chance points; continue to the first decision point.

→ **Drawing rules:** 1. begins with a **decision node (rectangle)**; chance node = circle; 2. alternatives shown as **straight lines** from the decision node; 3. drawn left to right, rectangles and circles **numbered sequentially**; 4. add values and probabilities for each branch.

→ **Evaluation (right to left):** (a) **EMV at a chance node** = Σ (value × probability) of its branches; (b) value at a **decision node** = **highest** of its branches' expected values.

→ **Joint probability** of a path = product of probabilities along it (app-written from ICAI TYK P20 figures, e.g. 0.4 × 0.2 = 0.08); expected NPV = Σ path NPV × joint probability.

ICAI ·

Illus 11 (L&R, ₹ lacs, 10%): pilot 2.40; success 50%. Decision C (success): invest 20 for perpetual 4 → 4/0.10 − 20 = **+20** → invest. Decision D (failure): invest 12 for perpetual 1 → 10 − 12 = **−2** → don't invest. Outcome B: EMV = 20 × 0.5 + 0 × 0.5 = **10**. At A: spend 2,40,000 to test for EMV 10 lacs → **undertake the test**.

TYK P20: outlay 80,000, 10%. Path NPVs: −14,726; −8,118; 1,794; 7,580; 15,840; 24,100; joint probabilities 0.08, 0.12, 0.20, 0.24, 0.30, 0.06 (e.g. 0.4 × 0.2). Expected NPV **₹ 6,223.76**. Worst: −14,726 at 8% (path 1); best: 24,100 at 6% (path 6). Positive ENPV → **accept**.

Note (app-written): Illus 11's diagram numbers nodes 1, 2, 3 and X, Y, Z, but the evaluation text refers to points A, B, C, D (SM p.36–37).

5. Replacement decision

→ One of the most important classes of capital budgeting; funds are limited, so investment in long-term assets must be budgeted. Two types: 5.1 replacement of existing machine; 5.2 optimum replacement cycle.

5.1 Replacement of existing machine

→ Replace with newer version or different machine of the same function — usually to **maintain existing operations**; profitability may change via expenses.

→ Use **incremental** cash flows: cash flows of the old asset disappear (e.g. new sales 1,00,000 vs old 75,000 → incremental 25,000); sale value of old asset reduces the effective price of the new (old sold for 3,50,000 → new costs 3,50,000 less).

→ Include **depreciation** (affects taxes) and **taxes** — both change.

Step	Formula / rule	Illus 12 (Roby's cube)
I(a) Tax on sale of old	(Book value of old − Market value of old) × Tax rate = tax payable / saving	[(25,000 − 5 × 2,500) − 5,000] × 0.30 = **2,250** saving
I(b) Net cash outflow	Cost of new − [tax saving (or − tax payable) + market value of old]	50,000 − [2,250 + 5,000] = **42,750**
II Change in annual CF	[(Δ Sales ± Δ Operating costs) − Δ Depreciation](1 − t) + Δ Depreciation	[1,00,000 × 0.1 + 5,000 − (49,000/5 − 25,000/10)] (0.70) + (9,800 − 2,500) = **12,690**
III PV of benefits	PV of yearly CF + PV of salvage of new	12,690 × PVIFA(10%, 5) + 1,000 × PVIF(10%, 5) = **48,723**
IV NPV · V Rule	PV benefits − PV costs; accept if PV benefits > PV costs	48,723 − 42,750 = **5,973** → replace

Note (app-written): Illus 12 says sales rise 10% on ₹ 10,00,000 (i.e. ₹ 1,00,000), but ICAI's working takes ₹ 1,00,000 × 0.1 = ₹ 10,000 as the change in sales; ₹ 12,690 and NPV ₹ 5,973 rest on ₹ 10,000 (SM p.39).

ICAI ·

TYK P23 (upgrade vs new): old BV nil, sells 50,000 → net of 40% tax 30,000. Upgrade outlay 10,00,000 (dep 2,00,000); new 20,50,000 − 30,000 = 20,20,000 (dep 4,10,000). CFAT = EAT + dep; incremental over old (old EAT = CFAT): upgrade 2,50,000 / 2,50,000 / 2,30,000 / 2,30,000 / 2,40,000 → PV 8,08,680 → NPV **−1,91,320**; new 5,10,000 / 5,10,000 / 5,20,000 / 5,30,000 / 5,50,000 → PV 17,47,930 → NPV **−2,72,070** → **continue with old machine**.

5.2 Optimum replacement cycle

→ When the asset is replaced in a **continuous cycle**, simple NPV must be modified: NPVs of different periods are **not comparable** → use **Equivalent Annual Cost (EAC)**. Choose the cycle with the **lowest EAC**.

→ Assumption: as the asset ages, **efficiency falls** → operating cost rises and resale value falls.

Formula	Terms / logic	SM example figures
EAC = PV of Cash Outflow ÷ PVAF	PV outflow for keeping the asset n years = cost + PV of running costs − PV of resale at n. Divide by annuity factor for the same n → cost per year.	Illus 13: 3,09,100 ÷ 0.909 = 3,40,044; 5,47,100 ÷ 1.735 = 3,15,331; 7,89,650 ÷ 2.486 = 3,17,639.

ICAI ·

Illus 13 (taxi): *cost 4,00,000, 10%. Keep 1 yr: −4,00,000 − 1,80,000(0.909) + 2,80,000(0.909) = −3,09,100. 2 yrs: −4,00,000 − 1,80,000 × 0.909 + 20,000 × 0.826 = −5,47,100 (yr 2: resale 2,30,000 − cost 2,10,000). 3 yrs: −4,00,000 − 1,80,000 × 0.909 − 2,10,000 × 0.826 − 70,000 × 0.751 = −7,89,650. Lowest EAC* ***3,15,331*** *→* ***replace every 2 years.***

TYK P26 (machine, 15%): *net cash flows include maintenance + repair (16,000; 22,000; 28,000; 36,000) less scrap. PV of costs: 1 yr −46,086; 2 yrs −72,402; 3 yrs −98,438; 4 yrs −1,24,968. EAC: 46,086 ÷ 0.8696 = 52,997; 72,402 ÷ 1.6257 = 44,536; 98,438 ÷ 2.2832 =* ***43,114****; 1,24,968 ÷ 2.855 = 43,772 →* ***3-year cycle.***

TYK P27 (bikes, 10%): *price 55,000. Net outflow after PV of resale: 1 yr (1,818); 2 yrs 44,039; 3 yrs 89,172. EAC: 53,182 ÷ 0.909 = 58,506; 99,039 ÷ 1.735 =* ***57,083****; 1,44,172 ÷ 2.486 = 57,993 →* ***replace every 2 years.***

TYK P24 (machine A vs B, real cash flows, 10%): *A: 1,50,000 + 40,000 × 2.486 = 2,49,440 ÷ 2.486 =* ***1,00,338****; B: 1,00,000 + 60,000 × 1.735 = 2,04,100 ÷ 1.735 =* ***1,17,637*** *→ buy* ***A*** *(lower equivalent annual outflow).*

TYK P22 (A & Co., replace vs overhaul, 15%, no tax): *EAC of new = [90,000 + 10,000 × 4.4873 − 20,000 × 0.3269] ÷ 4.4873 = 1,28,335 ÷ 4.4873 =* ***28,600****. PV of cost of replacing: immediately 11,400 (net inflow: salvage 40,000 − EAC 28,600); in 1 yr (11,832); 2 yrs (34,102); 3 yrs (55,799); 4 yrs (82,799) →* ***replace immediately****. Alternative table: 41,652 / 43,739 / 47,633 / 53,321 / 66,414 → same answer.*

TYK P25 (replace now or after a year, 10%): *NPV of new = 80,000 × 2.486 − 1,50,000 = 48,880. Now: 80,000 + 48,880 =* ***1,28,880****. After 1 yr: (40,000 + 70,000 + 48,880) ÷ 1.1 = 1,58,880 ÷ 1.1 =* ***1,44,436*** *→* ***replace after 1 year.***

6. Adjusted present value (APV)

- → *Instead of adjusting the discount rate (WACC) for risk, APV* ***separates the investment decision from the financing decision.***
- → ***APV = Base case NPV + PV of tax benefit on interest.***
- → ***Base case NPV*** *— discount at* ***unlevered cost of equity*** *(all-equity firm); computing it is covered in the Business Valuation chapter.*
- → ***PV of tax benefit on interest*** *— adjusts for how the project is financed → broader view including the benefit of using more debt.*

Exam steps — inflation NPV: *check if k is nominal or real and if CFs are nominal or real → match them (Nominal Cash Flow = (1 + Inflation Rate)^t × Real Cash Flow, or Real Discount Rate = (1 + Nominal Discount Rate)/(1 + Inflation Rate) − 1) → if revenue and cost inflation differ, compound each separately → (R − C)(1 − T) + DT (shield not inflated) → discount → NPV.*

Exam steps — ENPV, SD, CV: *ENCF = Σ NCF_i × P_i (each year) → discount → ENPV → σ^2 = Σ $(NCF_j - ENCF)^2$ P_j → σ (multi-year: Hillier $\sigma^2 = \Sigma (1+r)^{-2i} \sigma_i^2$) → CV = σ ÷ mean → same mean: lower σ; different mean: lower CV.*

Exam steps — RADR / CE: *RADR = Risk free rate + Risk premium (or $IR_f + (k_o - IR_f)$ Risk Factor; or table by CV) → discount unadjusted CF. CE: $NCF_t \times \alpha_t$ → discount at* ***risk-free rate*** *only → NPV; lower $\Sigma\alpha$ = riskier = higher RADR.*

Exam steps — sensitivity: base NPV → change one variable (given % adverse) → revised NPV → % change = (base − revised) ÷ base → highest % = most sensitive. Or NPV-zero method: % = NPV ÷ PV of that variable → lowest % = most sensitive. k: (IRR − X) ÷ X.

Exam steps — scenario / best-worst: NPV for worst, most likely, best → expected NPV with probabilities → σ, CV → worst-case probability: dependent = year-1 probability, independent = that probability^n → compare with acceptable risk level.

Exam steps — decision tree: draw left to right (□ decision, ○ chance) → path NPVs → joint probability = product along path → roll back right to left: chance node = Σ value × p; decision node = highest → decide at first node.

Exam steps — replacement: tax on sale of old = (BV − MV) × t → net outflow = cost of new − (MV + tax saving) → Δ CF = [(ΔS ± ΔC) − ΔD](1 − t) + ΔD → PV of Δ CF + PV of new salvage → NPV → accept if > 0.

Exam steps — optimum replacement cycle: for each life n: PV of cost + running costs − PV resale at n → EAC = PV ÷ PVAF(k, n) → lowest EAC = optimum cycle. Replace now vs later: compare PV of each timing with new asset's NPV included.

③ Advanced Capital Budgeting Decisions — what ICAI asked (Sep 2025, Jan 2026, May 2026)

12 marks *set across the last three attempts (3.4% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	MCQ CS-I 1-3	6	Sensitivity analysis of NPV: % change in NPV when initial cost, annual cash inflow, or cost of capital is varied adversely by 10% (MCQ)
Jan 2026	MCQ CS-V 13-15	6	Choice between a sure product and an uncertain one: profit from contribution minus fixed cost, and expected value of profit using a probability distribution (MCQ)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Write short note on Certainty Equivalent Approach.

ICAI answer

1. This approach recognizes risk in capital budgeting analysis by adjusting estimated cash flows and employs risk free rate to discount the adjusted cash-flows. Under this method, the expected cash flows of the project are converted to equivalent riskless amounts. The greater the risk of an expected cash flow, the smaller the certainty equivalent values for receipts and longer the CE value for payment. This approach is superior to the risk adjusted discounted approach as it can measure risk more accurately.

This is yet another approach for dealing with risk in capital budgeting to reduce the forecasts of cash flows to some conservative levels. In certainty Equivalent approach we incorporate risk to adjust the cash flows of a proposal so as to reflect the risk element. The certainty Equivalent approach adjusts future cash flows rather than discount rates. This approach explicitly recognizes risk, but the procedure for reducing the forecasts of cash flows is implicit and likely to be inconsistent from one investment to another.

Theory Q2

2. What is the sensitivity analysis in Capital Budgeting?

ICAI answer

2. Sensitivity analysis is used in Capital budgeting for more precisely measuring the risk. It helps in assessing information as to how sensitive are the estimated parameters of the project such as cash flows, discount rate, and the project life to the estimation errors. Future being always uncertain and estimations are always subject to error, sensitivity analysis takes care of estimation errors by using a number of possible outcomes in evaluating a project. The

methodology adopted in sensitivity analysis is to evaluate a project by using a number of estimated cash flows so as to provide to the decision maker an insight into the variability of outcome. Thus, it is a technique of risk analysis which studies the responsiveness of a criterion of merit like NPV or IRR to variation in underlying factors like selling price, quantity sold, returns from an investment etc.

Sensitivity analysis answers questions like,

(i) What happens to the present value (or some other criterion of merit) if flows are, say ₹ 50,000 than the expected ₹ 80,000?

(ii) What will happen to NPV if the economic life of the project is only 3 years rather than expected 5 years?

Therefore, wherever there is an uncertainty, of whatever type, the sensitivity analysis plays a crucial role. However, it should not be viewed as the method to remove the risk or uncertainty, it is only a tool to analyse and measure the risk and uncertainty. In terms of capital budgeting the possible cash flows are based on three assumptions:

(a) Cash flows may be worst (pessimistic)

(b) Cash flows may be most likely.

(c) Cash flows may be most optimistic.

Sensitivity analysis involves three steps

(1) Identification of all those variables having an influence on the project's NPV or IRR.

(2) Definition of the underlying quantitative relationship among the variables.

(3) Analysis of the impact of the changes in each of the variables on the NPV of the project.

The decision maker, in sensitivity analysis always asks himself the question – what if?

Theory Q3

3. Write a note on project appraisal under inflationary conditions.

ICAI answer

3. Project Appraisal normally involves feasibility evaluation from technical, commercial, economic and financial aspects. It is generally an exercise in measurement and analysis of cash flows expected to occur over the life of the project. The project cash outflows usually occur initially and inflows come in the future.

During inflationary conditions, the project cost increases on all heads viz. labour, raw material, fixed assets such as equipments, plant and machinery, building material, remuneration of technicians and managerial personnel etc. Beside this, inflationary conditions

erode purchasing power of consumers and affect the demand pattern. Thus, not only cost of production but also the projected statement of profitability and cash flows are affected by the change in demand pattern. Even financial institutions and banks may revise their lending rates resulting in escalation in financing cost during inflationary conditions. Under such circumstances, project appraisal has to be done generally keeping in view the following guidelines which are usually followed by government agencies, banks and financial institutions.

(i) It is always advisable to make provisions for cost escalation on all heads of cost, keeping in view the rate of inflation during likely period of delay in project implementation.

(ii) The various sources of finance should be carefully scrutinized with reference to probable revision in the rate of interest by the lenders and the revision which could be affected in the interest-bearing securities to be issued. All these factors will push up the cost of funds for the organization.

(iii) Adjustments should be made in profitability and cash flow projections to take care of the inflationary pressures affecting future projections.

(iv) It is also advisable to examine the financial viability of the project at the revised rates and assess the same with reference to economic justification of the project. The appropriate measure for this aspect is the economic rate of return for the project which will equate the present value of capital expenditures to net cash flows over the life of the projects. The rate of return should be acceptable which also accommodates the rate of inflation per annum.

(v) In an inflationary situation, projects having early payback periods should be preferred because projects with long payback period are riskier.

Under conditions of inflation, the project cost estimates that are relevant for a future date will suffer escalation. Inflationary conditions will tend to initiate the measurement of future cash flows. Either of the following two approaches may be used while appraising projects under such conditions:

(a) Adjust each year's cash flows to an inflation index, recognizing selling price increases and cost increases annually; or

(b) Adjust the 'Acceptance Rate' (cut-off) suitably retaining cash flow projections at current price levels.

An example of approach (ii) above can be as follows:

Normal Acceptance Rate	:	15.0%
Expected Annual Inflation	:	5.0%
Adjusted Discount Rate	:	15.0 × 1.05 or 15.75%

It must be noted that measurement of inflation has no standard approach nor is easy. This makes the job of appraisal a difficult one under such conditions.

Theory Q4

4. Explain the steps involved in Simulation Analysis.

ICAI answer: "Please refer paragraph 4.3.3" – SM paragraph 4.3.3 reproduced

4.3.3 Simulation Analysis (Monte Carlo)

Simulation is the exact replica of the actual situation. To simulate an actual situation, a model shall be prepared. The simulation Analysis is a technique, in which infinite calculations are made to obtain the possible outcomes and probabilities for any given action.

Monte Carlo simulation ties together sensitivities and probability distributions. The method came out of the work of first nuclear bomb and was so named because it was based on mathematics of Casino gambling. Fundamental appeal of this analysis is that it provides decision makers with a probability distribution of NPVs rather than a single point estimates of the expected NPV.

This analysis starts with carrying out a simulation exercise to model the investment project. It involves identifying the key factors affecting the project and their inter relationships. It involves modelling of cash flows to reveal the key factors influencing both cash receipt and payments and their inter relationship.

This analysis specifies a range for a probability distribution of potential outcomes for each of model's assumptions.

4.3.3.1 Steps for Simulation Analysis:

1. Modelling the project: The model shows the relationship of NPV with parameters and exogenous variables. (Parameters are input variables specified by decision maker and held constant over all simulation runs. Exogenous variables are input variables, which are stochastic in nature and outside the control of the decision maker).
2. Specify values of parameters and probability distributions of exogenous variables.
3. Select a value at random from probability distribution of each of the exogenous variables.
4. Determine NPV corresponding to the randomly generated value of exogenous variables and pre-specified parameter variables.
5. Repeat steps (3) & (4) a large number of times to get a large number of simulated NPVs.
6. Plot probability distribution of NPVs and compute a mean and Standard Deviation of returns to gauge the project's level of risk.

Example: Uncertainty associated with two aspects of the project: Annual Net Cash Flow & Life of the project. NPV model for the project is

$$\sum_{t=1}^{n} [CF_t/(1+i)^t] - I$$

Where i→ Risk free interest rate, I→ initial investment are parameters, CF = Annual Cash Flow

With i = 10%, I = ₹ 1,30,000, CF_t & n stochastic exogenous variables with the following distribution will be as under:

Annual Cash Flow		Project Life	
Value (₹)	Probability	Value (Year)	Probability
10,000	0.02	3	0.05
15,000	0.03	4	0.10
20,000	0.15	5	0.30
25,000	0.15	6	0.25
30,000	0.30	7	0.15
35,000	0.20	8	0.10
40,000	0.15	9	0.03
		10	0.02

Ten manual simulation runs are performed for the project. To perform this operation, values are generated at random for the two exogenous variables viz., Annual Cash Flow and Project Life. For this purpose, we take following steps

(1) set up correspondence between values of exogenous variables and random numbers

(2) choose some random number generating device.

Correspondence between Values of Exogenous Variables and two Digit Random Numbers:

Annual Cash Flow				Project Life			
Value (₹)	Probability	Cumulative Probability	Two Digit Random No.	Value (Year)	Probability	Cumulative Probability	Two Digit Random No.
10,000	0.02	0.02	00 – 01	3	0.05	0.05	00 – 04
15,000	0.03	0.05	02 – 04	4	0.10	0.15	05 – 14
20,000	0.15	0.20	05 – 19	5	0.30	0.45	15 – 44
25,000	0.15	0.35	20 – 34	6	0.25	0.70	45 – 69
30,000	0.30	0.65	35 – 64	7	0.15	0.85	70 – 84

35,000	0.20	0.85	65 – 84	8	0.10	0.95	85 – 94
40,000	0.15	1.00	85 - 99	9	0.03	0.98	95 – 97
				10	0.02	1.00	98 - 99

Random Number

53479	81115	98036	12217	59526
97344	70328	58116	91964	26240
66023	38277	74523	71118	84892
99776	75723	03172	43112	83086
30176	48979	92153	38416	42436
81874	83339	14988	99937	13213
19839	90630	71863	95053	55532
09337	33435	53869	52769	18801
31151	58295	40823	41330	21093
67619	52515	03037	81699	17106

For random numbers, we can begin from any-where taking at random from the table and read any pair of adjacent columns, column/row wise. For the first simulation run we need two digit random numbers (1) For Annual Cash Flow (2) For Project Life. The numbers are 53 & 97 and corresponding value of Annual Cash Flow and Project Life are ₹ 3,000 and 9 years respectively.

Simulation Results

Annual Cash Flow			Project Life			
Run	Random No.	Corres. Value of Annual Cash Flow (1)	Random No.	Corres. Value of Project Life	PVAF @ 10% (2)	NPV (1)x(2) – 1,30,000
1	53	30,000	97	9	5.759	42,770
2	66	35,000	99	10	6.145	85,075
3	30	25,000	81	7	4.868	(8,300)
4	19	20,000	09	4	3.170	(66,600)
5	31	25,000	67	6	4.355	(21,125)
6	81	35,000	70	7	4.868	40,380
7	38	30,000	75	7	4.868	16,040
8	48	30,000	83	7	4.868	16,040
9	90	40,000	33	5	3.791	21,640
10	58	30,000	52	6	4.355	650

(1) We can predict all type of bad market situation beforehand.

(2) Handle problems characterised by:

(a) numerous exogenous variables following any kind of distribution.

(b) complex inter-relationships among parameters, exogenous variables and endogenous variables. Such problems defy capabilities of analytical methods.

(c) compels decision maker to explicitly consider the inter-dependencies and uncertainties featuring the project.

4.3.3.3 Shortcomings

(1) Difficult to model the project and specify probability distribution of exogenous variables.

(2) Simulation is inherently imprecise. Provides rough approximation of probability distribution of NPV Due to its imprecision, simulation probability distribution may be misleading when a tail of distribution is critical.

(3) Realistic simulation model being likely to be complex would probably be constructed by management expert and not by the decision maker. Decision maker lacking understanding of the model may not use it.

(4) Determine NPV in simulation run, risk free discount rate is used. It is done to avoid pre-judging risk, which is reflected in the dispersion of the distribution of N.P.V. This derived measure of NPV takes a different meaning from its original value, and, therefore, is difficult to interpret.

ICAI answer: "Please refer paragraph 4.3.3" – SM paragraph 4.3.3 reproduced

4.3.3 Simulation Analysis (Monte Carlo)

Simulation is the exact replica of the actual situation. To simulate an actual situation, a model shall be prepared. The simulation Analysis is a technique, in which infinite calculations are made to obtain the possible outcomes and probabilities for any given action.

Monte Carlo simulation ties together sensitivities and probability distributions. The method came out of the work of first nuclear bomb and was so named because it was based on mathematics of Casino gambling. Fundamental appeal of this analysis is that it provides decision makers with a probability distribution of NPVs rather than a single point estimates of the expected NPV.

This analysis starts with carrying out a simulation exercise to model the investment project. It involves identifying the key factors affecting the project and their inter relationships. It involves modelling of cash flows to reveal the key factors influencing both cash receipt and payments and their inter relationship.

This analysis specifies a range for a probability distribution of potential outcomes for each of model's assumptions.

4.3.3.1 Steps for Simulation Analysis:

1. Modelling the project: The model shows the relationship of NPV with parameters and exogenous variables. (Parameters are input variables specified by decision maker and held constant over all simulation runs. Exogenous variables are input variables, which are stochastic in nature and outside the control of the decision maker).
2. Specify values of parameters and probability distributions of exogenous variables.
3. Select a value at random from probability distribution of each of the exogenous variables.
4. Determine NPV corresponding to the randomly generated value of exogenous variables and pre-specified parameter variables.
5. Repeat steps (3) & (4) a large number of times to get a large number of simulated NPVs.
6. Plot probability distribution of NPVs and compute a mean and Standard Deviation of returns to gauge the project's level of risk.

Example: Uncertainty associated with two aspects of the project: Annual Net Cash Flow & Life of the project. NPV model for the project is

$$\sum_{t=1}^{n} [CF_t/(1+i)^t] - I$$

Where i→ Risk free interest rate, I→ initial investment are parameters, CF = Annual Cash Flow

With i = 10%, I = ₹ 1,30,000, CF_t & n stochastic exogenous variables with the following distribution will be as under:

Annual Cash Flow		Project Life	
Value (₹)	Probability	Value (Year)	Probability
10,000	0.02	3	0.05
15,000	0.03	4	0.10
20,000	0.15	5	0.30
25,000	0.15	6	0.25
30,000	0.30	7	0.15
35,000	0.20	8	0.10
40,000	0.15	9	0.03
		10	0.02

Ten manual simulation runs are performed for the project. To perform this operation, values are generated at random for the two exogenous variables viz., Annual Cash Flow and Project Life. For this purpose, we take following steps

(1) set up correspondence between values of exogenous variables and random numbers

(2) choose some random number generating device.

Correspondence between Values of Exogenous Variables and two Digit Random Numbers:

Annual Cash Flow				Project Life			
Value (₹)	Probability	Cumulative Probability	Two Digit Random No.	Value (Year)	Probability	Cumulative Probability	Two Digit Random No.
10,000	0.02	0.02	00 – 01	3	0.05	0.05	00 – 04
15,000	0.03	0.05	02 – 04	4	0.10	0.15	05 – 14
20,000	0.15	0.20	05 – 19	5	0.30	0.45	15 – 44
25,000	0.15	0.35	20 – 34	6	0.25	0.70	45 – 69
30,000	0.30	0.65	35 – 64	7	0.15	0.85	70 – 84

35,000	0.20	0.85	65 – 84	8	0.10	0.95	85 – 94
40,000	0.15	1.00	85 - 99	9	0.03	0.98	95 – 97
				10	0.02	1.00	98 - 99

Random Number

53479	81115	98036	12217	59526
97344	70328	58116	91964	26240
66023	38277	74523	71118	84892
99776	75723	03172	43112	83086
30176	48979	92153	38416	42436
81874	83339	14988	99937	13213
19839	90630	71863	95053	55532
09337	33435	53869	52769	18801
31151	58295	40823	41330	21093
67619	52515	03037	81699	17106

For random numbers, we can begin from any-where taking at random from the table and read any pair of adjacent columns, column/row wise. For the first simulation run we need two digit random numbers (1) For Annual Cash Flow (2) For Project Life. The numbers are 53 & 97 and corresponding value of Annual Cash Flow and Project Life are ₹ 3,000 and 9 years respectively.

Simulation Results

Annual Cash Flow			Project Life			
Run	Random No.	Corres. Value of Annual Cash Flow (1)	Random No.	Corres. Value of Project Life	PVAF @ 10% (2)	NPV (1)x(2) – 1,30,000
1	53	30,000	97	9	5.759	42,770
2	66	35,000	99	10	6.145	85,075
3	30	25,000	81	7	4.868	(8,300)
4	19	20,000	09	4	3.170	(66,600)
5	31	25,000	67	6	4.355	(21,125)
6	81	35,000	70	7	4.868	40,380
7	38	30,000	75	7	4.868	16,040
8	48	30,000	83	7	4.868	16,040
9	90	40,000	33	5	3.791	21,640
10	58	30,000	52	6	4.355	650

(1) We can predict all type of bad market situation beforehand.

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(b) complex inter-relationships among parameters, exogenous variables and endogenous variables. Such problems defy capabilities of analytical methods.

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(1) Difficult to model the project and specify probability distribution of exogenous variables.

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(4) Determine NPV in simulation run, risk free discount rate is used. It is done to avoid pre-judging risk, which is reflected in the dispersion of the distribution of N.P.V. This derived measure of NPV takes a different meaning from its original value, and, therefore, is difficult to interpret.

① SM illustrations with ICAI solutions

I. Current Trends In Capital Budgeting

SM Ch 3 · Illustration 1 (under 1.1 Impact of Inflation on Capital Budgeting Decisions)

Illustration 1

Determine NPV of the project with the following information:

Initial Outlay of project	₹ 40,000
Annual Cash Flow from the Project (Without inflation)	₹ 15,000
Useful life	4 years
Cost of Capital (Including inflation premium of 10%)	12%

Solution — ICAI

Solution

Annual Cash Flow of project is ₹ 15,000.

It would be inconsistent to discount these real cash flows at 12% (nominal rate of return).

There are two alternatives:

(i) Either to restate the cash flow in nominal term and discount at 12% or

(ii) Restate the discount rate in real terms and use this to discount the real cash flows.

NPV using (i) approach

Since inflation rate is 10% a year, real cash flows may be stated in nominal cash flows as follows:

Nominal Cash Flow = (1 + Inflation Rate) Real Cash Flows

Year	Real Cash Flows	Nominal Cash flows
1	15000	15,000 × 1.10 = 16,500
2	15,000	15,000 × $(1.10)^2$ = 18,150
3	15,000	15,000 × $(1.10.)^3$ = 19,965
4	15,000	15,000 × $(1.10)^4$ = 21,962

NPV using nominal discounting rate 12%

$$\frac{16,500}{(1.12)}+\frac{18,150}{(1.12)^2}+\frac{19,965}{(1.12)^3}+\frac{21,962}{(1.12)^4}-40,000$$

= ₹ 14,732 + ₹ 14,469 + ₹ 14,211+ ₹ 13,957 – ₹ 40,000

= ₹ 17,369 (Approx)

NPV using (ii) approach

To compute NPV using (ii) approach, we shall need real discount rate, which shall be computed as follows:

$$\text{Real Discount Rate}=\frac{1+\text{Nominal Discount Rate}}{1+\text{Inflation Rate}}-1$$

$$\text{Real Discount Rate}=\frac{1+0.12}{1+0.10}-1=0.0182 \text{ i.e. } 1.8\%.$$

$$NPV = \sum_{t=1}^{n} cf_t - I_o$$

Where t = Time Period

cf_t = Annual Cash Flow

I_o = Initial Outlay

Accordingly NPV of the project

$$\frac{15,000}{(1.0182)} + \frac{15,000}{(1.0182)^2} + \frac{15,000}{(1.0182)^3} + \frac{15,000}{(1.0182)^4} - 40,000$$

= ₹ 14,732 + ₹ 14,469 + ₹ 14,210+ ₹ 13,956 – ₹ 40,000

= ₹ 57,367 – ₹ 40,000 = ₹17,367(Approx)

NPV based on consideration that inflation rate for revenue and cost are different shall be computed as follows:

$$N.P.V. = {}^{n}\Sigma_{t=1} [\{R_t (1+i_r) - C_t{}^{t}\Sigma_{r=1}(1+i_c)\} (1-T) + D_t T] / (1+k)^t - I_0$$

R_t→ revenues for the year 't' with no inflation.

i_r→ annual inflation rate in revenues for 'r th ' year.

C_t→ costs for year 't' with no inflation.

i_c→ annual inflation rate of costs for year 'r'.

T → tax rate.

D_t→ depreciation charge for year 't'.

I_0→ initial outlay.

k → cost of capital (with inflation premium).

SM Ch 3 · Illustration 2 (under 1.1 Impact of Inflation on Capital Budgeting Decisions)

Illustration 2

XYZ Ltd. requires ₹ 8,00,000 for a project. Useful life of project - 4 years. Salvage value - Nil. Depreciation Charge ₹ 2,00,000 p.a. Expected revenues & costs (excluding depreciation) ignoring inflation.

Year	1	2	3	4
Revenues	₹ 6,00,000	₹ 7,00,000	₹ 8,00,000	₹ 8,00,000
Costs	₹ 3,00,000	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000

Applicable Tax Rate is 60% and cost of capital is 10% (including inflation premium).

Calculate NPV of the project if inflation rates for revenues & costs are as follows:

Year	Revenues	Costs
1	10%	12%
2	9%	10%
3	8%	9%
4	7%	8%

Solution — ICAI

Solution

Computation of Annual Cash Flow

(i) Inflation adjusted Revenues

Year	Revenues (₹)	Revenues (Inflation Adjusted) (₹)
1	6,00,000	6,00,000(1.10) = 6,60,000
2	7,00,000	7,00,000(1.10)(1.09) = 8,39,300
3	8,00,000	8,00,000(1.10)(1.09)(1.08) = 10,35,936
4	8,00,000	8,00,000(1.10)(1.09)(1.08)(1.07) = 11,08,452

(ii) Inflation adjusted Costs

Year	Revenues (₹)	Revenues (Inflation Adjusted) (₹)
1	3,00,000	3,00,000(1.12) = 3,36,000
2	4,00,000	4,00,000(1.12)(1.10) = 4,92,800
3	4,00,000	4,00,000(1.12)(1.10)(1.09) = 5,37,152
4	4,00,000	4,00,000(1.12)(1.10)(1.09)(1.08) = 5,80,124

(iii) Tax Benefit on Depreciation = ₹ 2,00,000 x 0.60 = ₹ 1,20,000

(iv) Net Profit after Tax

Year	Revenues (Inflation Adjusted) (₹)(1)	Costs (Inflation Adjusted) (₹)(2)	Net Profit (₹) (3) = (1) - (2)	Tax (₹) (4) = 60% of (3)	Net after Profit (₹) (3) - (4)
1	6,60,000	3,36,000	3,24,000	1,94,400	1,29,600
2	8,39,300	4,92,800	3,46,500	2,07,900	1,38,600
3	10,35,936	5,37,152	4,98,784	2,99,270	1,99,514
4	11,08,452	5,80,124	5,28,328	3,16,997	2,11,331

(iv) Present Value of Cash Inflows

Year	Net after Profit (₹)	Tax Benefit on Depreciation (₹)	Cash Inflow (₹)	PVF@ 10%	PV (₹)
1	1,29,600	1,20,000	2,49,600	0.909	2,26,886
2	1,38,600	1,20,000	2,58,600	0.826	2,13,604
3	1,99,514	1,20,000	3,19,514	0.751	2,39,955
4	2,11,331	1,20,000	3,31,331	0.683	2,26,299
					9,06,744

NPV = ₹ 9,06,744 – ₹ 8,00,000 = ₹ 1,06,744

4. Methods Of Incorporating Risk In Capital

SM Ch 3 · Illustration 3 (under 4.1.1 Probability)

Illustration 3

Possible net cash flows of Projects A and B at the end of first year and their probabilities are given below. Discount rate is 10 per cent. For both the projects, initial investment is ₹ 10,000. Calculate the expected net present value for each project. State which project is preferable?

Possible Event	Project A		Project B	
	Cash Flow (₹)	Probability	Cash Flow (₹)	Probability
A	8,000	0.10	24,000	0.10
B	10,000	0.20	20,000	0.15
C	12,000	0.40	16,000	0.50
D	14,000	0.20	12,000	0.15
E	16,000	0.10	8 ,000	0.10

Solution — ICAI

Solution

Calculation of Expected Value for Project A and Project B

	Project A			Project B		
Possible Event	Cash Flow (₹)	Probability	Expected Value (₹)	Cash Flow (₹)	Probability	Expected Value (₹)
A	8,000	0.10	800	24,000	0.10	2,400
B	10,000	0.20	2,000	20,000	0.15	3,000
C	12,000	0.40	4,800	16,000	0.50	8,000
D	14,000	0.20	2,800	12,000	0.15	1,800
E	16,000	0.10	1,600	8,000	0.10	800
ENCF			12,000			16,000

The Net Present Value for Project A is (0.909 × ₹ 12,000 – ₹ 10,000) = ₹ 908

The Net Present Value for Project B is (0.909 × ₹ 16,000 – ₹ 10,000) = ₹ 4,544.

(b) Expected Net Present Value- Multiple period

Let us understand the calculation of Expected Net Present Value (ENPV) for multiple periods through an illustration as follows:

SM Ch 3 · Illustration 4 (under 4.1.1 Probability)

Illustration 4

Probabilities for net cash flows for 3 years of a project are as follows:

Year 1		Year 2		Year 3	
Cash Flow (₹)	Probability	Cash Flow (₹)	Probability	Cash Flow (₹)	Probability
2,000	0.1	2,000	0.2	2,000	0.3
4,000	0.2	4,000	0.3	4,000	0.4
6,000	0.3	6,000	0.4	6,000	0.2
8,000	0.4	8,000	0.1	8,000	0.1

Calculate the expected net present value of the project using 10 per cent discount rate if the Initial Investment of the project is ₹ 10,000.

Solution — ICAI

Solution

Calculation of Expected Value

Year 1			Year 2			Year 3		
Cash Flow (₹)	Prob.	Expected Value (₹)	Cash Flow (₹)	Prob.	Expected Value (₹)	Cash Flow (₹)	Prob.	Expected Value (₹)
2,000	0.1	200	2,000	0.2	400	2,000	0.3	600
4,000	0.2	800	4,000	0.3	1200	4,000	0.4	1,600
6,000	0.3	1,800	6,000	0.4	2400	6,000	0.2	1,200
8,000	0.4	3,200	8,000	0.1	800	8,000	0.1	800
ENCF		6,000			4,800			4,200

The present value of the expected value of cash flow at 10 per cent discount rate has been determined as follows:

Present Value of cash flow $= \frac{ENCF_1}{(1+k)^1} + \frac{ENCF_2}{(1+k)^2} + \frac{ENCF_3}{(1+k)^3}$

$= \frac{6,000}{(1.1)} + \frac{4,800}{(1.1)^2} + \frac{4,200}{(1.1)^3}$

$= (6,000 \times 0.909) + (4,800 \times 0.826) + (4,200 \times 0.751)$

$= ₹ 12,573$

Expected Net Present value = Present Value of cash flow - Initial Investment

= ₹ 12,573 – ₹ 10,000 = ₹ 2,573.

SM Ch 3 · Illustration 5 (under 4.1.3 Standard Deviation)

Illustration 5

Calculate Variance and Standard Deviation of Project A and Project B on the basis of following information:

Possible Event	Project A		Project B	
	Cash Flow (₹)	Probability	Cash Flow (₹)	Probability
A	8,000	0.10	24,000	0.10
B	10,000	0.20	20,000	0.15
C	12,000	0.40	16,000	0.50
D	14,000	0.20	12,000	0.15
E	16,000	0.10	8,000	0.10

Solution — ICAI

Solution

Calculation of Expected Value for Project A and Project B

	Project A			Project B		
Possible Event	Cash Flow (₹)	Probability	Expected Value (₹)	Cash Flow (₹)	Probability	Expected Value (₹)
A	8,000	0.10	800	24,000	0.10	2,400
B	10,000	0.20	2,000	20,000	0.15	3,000
C	12,000	0.40	4,800	16,000	0.50	8,000
D	14,000	0.20	2,800	12,000	0.15	1,800
E	16,000	0.10	1,600	8,000	0.10	800
ENCF			12,000			16,000

Project A:

Variance ($\sigma 2$) = $(8,000 - 12,000)2 \times (0.1) + (10,000 - 12,000)2 \times (0.2) + (12,000 - 12000)2 \times (0.4) + (14,000 - 12,000)2 \times (0.2) + (16000 - 12,000)2 \times (0.1)$

= 16,00,000 + 8,00,000 + 0 + 8,00,000 + 16,00,000 = 48,00,000

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)}$ = $\sqrt{48,00,000}$ = 2,190.90

Project B:

Variance($\sigma 2$) = $(24,000 - 16,000)2 \times (0.1) + (20,000 - 16,000)2 \times (0.15) + (16,000 - 16,000)2 \times (0.5) + (12,000 - 16,000)2 \times (0.15) + (8,000 - 16,000)2 \times (0.1)$

= 64,00,000 + 24,00,000 + 0 + 24,00,000 + 64,00,000 = 1,76,00,000

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)}$ = $\sqrt{1,76,00,000}$ = 4195.23

SM Ch 3 · Illustration 6 (under 4.1.4 The Coefficient of Variation)

Illustration 6

Calculate Coefficient of Variation of Project A and Project B based on the following information:

Possible Event	Project A		Project B	
	Cash Flow (₹)	Probability	Cash Flow (₹)	Probability
A	10000	0.10	26,000	0.10
B	12,000	0.20	22,000	0.15
C	14,000	0.40	18,000	0.50
D	16,000	0.20	14,000	0.15
E	18,000	0.10	10,000	0.10

Solution — ICAI

Solution

Calculation of Expected Value for Project A and Project B

	Project A			Project B		
Possible Event	Cash Flow (₹)	Probability	Expected Value (₹)	Cash Flow (₹)	Probability	Expected Value (₹)
A	10,000	0.10	1,000	26,000	0.10	2,600
B	12,000	0.20	2,400	22,000	0.15	3,300
C	14,000	0.40	5,600	18,000	0.50	9,000
D	16,000	0.20	3,200	14,000	0.15	2,100
E	18,000	0.10	1,800	10,000	0.10	1,000
ENCF			14,000			18,000

Project A

Variance (σ^2) = $(10,000 - 14,000)^2 \times (0.1) + (12,000 - 14,000)^2 \times (0.2) + (14,000 - 14000)^2 \times (0.4) + (16,000 - 14,000)^2 \times (0.2) + (18000 - 14,000)^2 \times (0.1)$

= 16,00,000 + 8,00,000 + 0 + 8,00,000 + 16,00,000 = 48,00,000

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{48,00,000} = 2,190.90$

Project B:

Variance(σ^2) = $(26,000 - 18,000)^2 \times (0.1) + (22,000 - 18,000)^2 \times (0.15) + (18,000 - 18,000)^2 \times (0.5) + (14,000 - 18,000)^2 \times (0.15) + (10,000 - 18,000)^2 \times (0.1)$

= 64,00,000 + 24,00,000 + 0 + 24,00,000 + 64,00,000 = 1,76,00,000

Standard Deviation (σ) = $\sqrt{\text{Variance}(\sigma^2)} = \sqrt{1,76,00,000} = 4195.23$

Projects	Coefficient of variation	Risk	Expected Value
A	$\frac{2190.90}{14000} = 0.1565$	Less	Less
B	$\frac{4195.23}{18000} = 0.2331$	More	More

In project A, risk per rupee of cash flow is ₹ 0.16 while in project B, it is ₹ 0.23. Therefore, Project A is better than Project B.

SM Ch 3 · Illustration 7 (under 4.2.1 Risk Adjusted Discount Rate)

Illustration 7

An enterprise is investing ₹ 100 lakhs in a project. The risk-free rate of return is 7%. Risk premium expected by the Management is 7%. The life of the project is 5 years. Following are the cash flows that are estimated over the life of the project:

Year	Cash flows (₹ in lakhs)
1	25
2	60
3	75
4	80
5	65

Calculate Net Present Value of the project based on Risk free rate and also on the basis of Risks adjusted discount rate.

Solution — ICAI

Solution

The Present Value of the Cash Flows for all the years by discounting the cash flow at 7% is calculated as below:

Year	Cash flows (₹ in lakhs)	Discounting Factor @ 7%	Present value of Cash Flows (₹ In Lakhs)
1	25	0.935	23.38
2	60	0.873	52.38
3	75	0.816	61.20
4	80	0.763	61.04
5	65	0.713	46.35
Total of Present value of Cash flows			244.34
Less: Initial investment			100.00
Net Present Value (NPV)			**144.34**

Now, when the risk-free rate is 7% and the risk premium expected by the Management is 7%, then risk adjusted discount rate is 7% + 7% = 14%.

Discounting the above cash flows using the Risk Adjusted Discount Rate would be as below:

Year	Cash flows (₹ in Lakhs)	Discounting Factor @ 14%	Present Value of Cash Flows (₹ in lakhs)
1	25	0.877	21.93
2	60	0.769	46.14
3	75	0.675	50.63
4	80	0.592	47.36
5	65	0.519	33.74
Total of Present value of Cash flows			199.79
Less: Initial investment			100.00
Net present value (NPV)			99.79

Advantages of Risk-adjusted discount rate

(1) It is easy to understand.

(2) It incorporates risk premium in the discounting factor.

Limitations of Risk-adjusted discount rate

(1) Difficulty in finding risk premium and risk-adjusted discount rate.

(2) Though NPV can be calculated but it is not possible to calculate Standard Deviation of a given project.

SM Ch 3 · Illustration 8 (under 4.2.2 Certainty Equivalent (CE))

Illustration 8

If Investment proposal costs ₹ 45,00,000 and risk free rate is 5%, calculate net present value under certainty equivalent technique.

Year	Expected cash flow (₹)	Certainty Equivalent coefficient
1	10,00,000	0.90
2	15,00,000	0.85
3	20,00,000	0.82
4	25,00,000	0.78

Solution — ICAI

Solution

$$NPV = \frac{10,00,000\times(0.90)}{(1.05)} + \frac{15,00,000\times(0.85)}{(1.05)^2} + \frac{20,00,000\times(0.82)}{(1.05)^3} + \frac{25,00,000\times(0.78)}{(1.05)^4} - 45,00,000$$

= ₹ 5,34,570

Advantages of Certainty Equivalent Method

1. The certainty equivalent method is **simple and easy** to understand and apply.
2. It can **easily be calculated for different risk levels** applicable to different cash flows. For example, if in a particular year, a higher risk is associated with the cash flow, it can be easily adjusted and the NPV can be recalculated accordingly.

Disadvantages of Certainty Equivalent Method

1. There is **no objective** or mathematical method to estimate certainty equivalents. Certainty Equivalents are subjective and vary as per each individual's estimate.
2. Certainty equivalents are decided by the management based on their perception of risk. However, the **risk perception of the shareholders** who are the money lenders for the project is **ignored**. Hence, it is not used often in corporate decision making.

Risk-adjusted Discount Rate Vs. Certainty-Equivalent

Certainty Equivalent Method is superior to Risk Adjusted Discount Rate Method as it does not assume that risk increases with time at constant rate. Each year's Certainty Equivalent Coefficient is based on level of risk impacting its cash flow. Despite its soundness, it is not preferable like Risk Adjusted Discount Rate Method. It is difficult to specify a series of Certainty Equivalent Coefficients but simple to adjust discount rates.

SM Ch 3 · Illustration 9 (under 4.3.1 Sensitivity Analysis)

Illustration 9

X Ltd. is considering its new project with the following details:

Sr. No.	Particulars	Figures
1	Initial capital cost	₹ 400 Cr.
2	Annual unit sales	5 Cr.
3	Selling price per unit	₹ 100
4	Variable cost per unit	₹ 50
5	Fixed costs per year	₹ 50 Cr.
6	Discount Rate	6%

Required:

1. Calculate the NPV of the project.
2. Compute the impact on the project's NPV considering a 2.5 per cent adverse variance in each variable. Which variable is having maximum effect?

Consider Life of the project as 3 years.

Solution — ICAI

Solution

1. Calculation of Net Cash Inflow per year

	Particulars	Amount (₹)
A	Selling price per unit	100
B	Variable cost per unit	50
C	Contribution per unit (A - B)	50
D	Number of units sold per year	5 Cr.
E	Total Contribution (C × D)	₹ 250 Cr.
F	Fixed cost per year	₹ 50 Cr.
G	Net cash inflow per year (E - F)	₹ 200 Cr.

Calculation of Net Present Value (NPV) of the Project

Year	Year Cash Flow (₹ in Cr.)	PV factor @ 6%	Present Value (PV) (₹ in Cr.)
0	(400.00)	1.000	(400.00)
1	200.00	0.943	188.60
2	200.00	0.890	178.00
3	200.00	0.840	168.00
Net Present Value			134.60

Here, NPV represent the most likely outcomes and not the actual outcomes. The actual outcome can be lower or higher than the expected outcome.

2. Sensitivity Analysis considering 2.5 % Adverse Variance in each variable

	Particulars	Base	Initial capital cost increased to ₹ 410 crore	Selling Price per Unit Reduced to ₹ 97.5	Variable Cost Per Unit increased to ₹ 51.25	Fixed Cost Per Unit increased to ₹ 51.25	Units sold per year reduced to 4.875 crore
		(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
A	Selling price per unit	100	100	97.5	100	100	100

B	Variable cost per unit	50	50	50	51.25	50	50
C	Contribution per unit (A - B)	50	50	47.5	48.75	50	50
		(₹ in Cr.)	(₹ in Cr.)	(₹ in Cr.)	(₹ in Cr.)	(₹ in Cr.)	(₹ in Cr.)
D	Number of units sold per year (units in Crores)	5	5	5	5	5	4.875
E	Total Contribution (C × D)	250	250	237.5	243.75	250	243.75
F	Fixed cost per year	50	50	50	50	51.25	50
G	Net Cash Inflow per year (E - F)	200	200	187.5	193.75	198.75	193.75
H	PV of Net cash Inflow per year (G × 2.673)	534.60	534.60	501.19	517.89	531.26	517.89
I	Initial capital cost	400	410	400	400	400	400
J	NPV (H - I)	134.60	124.60	101.19	117.89	131.26	117.89
K	Percentage Change in NPV	-	-7.43%	-24.82%	-12.41%	-2.48%	-12.41%

The above table shows that by changing one variable at a time by 2.5% (adverse) while keeping the others constant, the impact in percentage terms on the NPV of the project can be calculated. Thus, the change in selling price has the maximum effect on the NPV by 24.82%.

Advantages of Sensitivity Analysis:

Following are the main advantages of Sensitivity Analysis:

(1) **Critical Issues**: This analysis identifies critical factors that impinge on a project's success or failure.

(2) **Simplicity**: It is a simple technique.

Disadvantage of Sensitivity Analysis

Following are the main disadvantages of Sensitivity Analysis:

(1) **Assumption of Independence**: This analysis assumes that all variables are independent i.e. they are not related to each other, which is unlikely in real life.

(2) **Ignore probability**: This analysis does not look to the probability of changes in the variables.

SM Ch 3 · Illustration 10 (under 4.3.2 Scenario Analysis)

Illustration 10

XYZ Ltd. is considering a project "A" with an initial outlay of ₹ 14,00,000 and the possible three cash inflow attached with the project as follows:

Particulars	Year 1	Year 2	Year 3
Worst case	450	400	700
Most likely	550	450	800
Best case	650	500	900

Assuming the cost of capital as 9%, determine NPV in each scenario. If XYZ Ltd is certain about the most likely result in first two years but uncertain about the third year's cash flow, analyze what will be the NPV expecting worst scenario in the third year.

Solution — ICAI

Solution

The possible outcomes will be as follows:

Year	PVF @ 9%	Worst Case		Most likely		Best case	
		Cash Flow	PV	Cash Flow	PV	Cash Flow	PV
		(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)	(₹ '000)
0	1	(1,400)	(1,400)	(1,400)	(1,400)	(1,400)	(1,400)
1	0.917	450	412.65	550	504.35	650	596.05
2	0.842	400	336.80	450	378.90	500	421.00
3	0.772	700	540.40	800	617.60	900	694.80
NPV			-110.15		100.85		311.85

If XYZ Ltd. is certain about the most likely result in first two years but uncertain about the third year's cash flow, then, NPV expecting worst case scenario is expected in the third year will be as follows:

$$= -₹\ 14,00,000 + \frac{₹\ 5,50,000}{(1+0.09)} + \frac{₹\ 4,50,000}{(1+0.09)^2} + \frac{₹\ 7,00,000}{(1+0.09)^3}$$

= − ₹ 14,00,000 + ₹ 5,04,587 + ₹ 3,78,756 + ₹ 5,40,528 = ₹ 23,871

Scenario Analysis Vs Sensitivity Analysis

- Sensitivity analysis and Scenario analysis both help to understand the impact of the change in input variable on the outcome of the project. However, there are certain basic differences between the two.
- Sensitivity analysis calculates the impact of the change of a single input variable on the outcome of the project viz., NPV or IRR. The sensitivity analysis thus enables to identify that single critical variable which can impact the outcome in a huge way and the range of outcomes of the project given the change in the input variable.
- Scenario analysis, on the other hand, is based on a scenario. The scenario may be recession or a boom wherein depending on the scenario, all input variables change. Scenario Analysis calculates the outcome of the project considering this scenario where the variables have changed simultaneously. Similarly, the outcome of the project would also be considered for the normal and recessionary situation. The variability in the outcome under the three different scenarios would help the management to assess the risk a project carries. Higher deviation in the outcome can be assessed as higher risk and lower to medium deviation can be assessed accordingly.

- Scenario analysis is far more complex than sensitivity analysis because in scenario analysis all inputs are changed simultaneously, considering the situation in hand while in sensitivity analysis, only one input is changed and others are kept constant.

SM Ch 3 · Illustration 11 (under 4.3.4. Decision Tree Analysis)

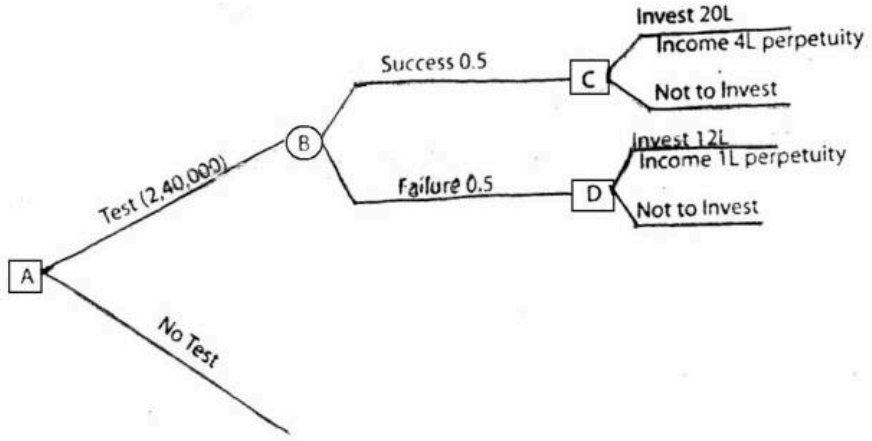
Illustration 11

L & R Limited wishes to develop new virus-cleaner software. The cost of the pilot project would be ₹ 2,40,000. Presently, the chances of the product being successfully launched on a commercial scale are rated at 50%. In case it does succeed. L&R can invest a sum of ₹ 20 lacs to market the product. Such an effort can generate perpetually, an annual net after tax cash income of ₹ 4 lacs. Even if the commercial launch fails, they can make an investment of a smaller amount of ₹ 12 lacs with the hope of gaining perpetually a sum of ₹ 1 lac. Evaluate the proposal, adopting decision tree approach. The discount rate is 10%.

Solution — ICAI

Solution

Decision tree diagram is given below:



Evaluation

At Decision Point C: The choice is between investing ₹ 20 lacs for a perpetual benefit of ₹ 4 lacs and not to invest. The preferred choice is to invest, since the capitalized value of benefit of ₹ 4 lacs (at 10%) adjusted for the investment of ₹ 20 lacs, yields a net benefit of ₹ 20 lacs.

At Decision Point D: The choice is between investing ₹ 12 lacs, for a similar perpetual benefit of ₹ 1 lac. and not to invest. Here the invested amount is greater than capitalized value of benefit at ₹ 10 lacs. There is a negative benefit of ₹ 2 lacs. Therefore, it would not be prudent to invest.

At Outcome Point B: Evaluation of EMV is as under (₹ in lacs).

Outcome	Amount (₹)	Probability	Result (₹)
Success	20.00	0.50	10.00
Failure	0.00	0.50	00.00
Net result			10.00

EMV at B is, therefore, ₹10 lacs.

At A: Decision is to be taken based on preferences between two alternatives. The first is to test, by investing ₹2,40,000 and reap a benefit of ₹ 10 lacs. The second is not to test, and thereby losing the opportunity of a possible gain.

The preferred choice is, therefore, investing a sum of ₹ 2,40,000 and undertaking the test.

5. Replacement Decision

SM Ch 3 · Illustration 12 (under 5.1 Replacement of Existing Machine)

Illustration 12

A Company named Roby's cube decided to replace the existing Computer system of their organisation. Original cost of old system was ₹ 25,000 and it was installed 5 years ago. Current market value of old system is ₹ 5,000. Depreciation of the old system was charged with life of 10 years with Estimated Salvage value as Nil. Depreciation of the new system will be charged with life over 5 years. Present cost of the new system is ₹ 50,000. Estimated Salvage value of the new system is ₹ 1,000. Estimated cost savings with new system is ₹ 5,000 per year. Increase in sales with new system is assumed at 10% per year based on original total sales of ₹ 10,00,000. Company follows straight line method of depreciation. Cost of capital of the company is 10% whereas tax rate is 30%.

Solution — ICAI

Solution

Step I. Net cash outflow (assumed at current time) [Present values of cost]:

a. (Book value of old system – market value of old system) × Tax Rate

= Tax payable/savings from sale

= [(₹ 25,000 – 5 × ₹ 2,500) – ₹ 5,000] × 0.30 = ₹ 7,500 × 0.30

= ₹ 2,250

b. Cost of new system – [Tax payable/savings from sale + Market value of old system] = Net cash outflow

Or, ₹ 50,000 – [₹ 2,250 + ₹ 5,000] = ₹42,750

Step II. Estimated change in cash flows per year if replacement decision is implemented.

Change in cash flow = [(Change in sales ± Change in operating costs)-Change in depreciation)] (1-tax rate) + Change in depreciation

= [₹ 1,00,000 × 0.1 + ₹ 5,000 – (₹ 49,000/5 – ₹ 25,000/10)] (1-0.30) + (₹ 49,000/5 – ₹ 25000/10)]

= ₹ 12,690

Step III. Present value of benefits = Present value of yearly cash flows + Present value of estimated salvage of new system

= ₹ 12,690 × PVIFA (10%, 5) + ₹ 1,000 × PVIF (10%, 5)

= ₹ 48,723

Step IV. Net present value = Present value of benefits - Present value of costs

= ₹ 48,723 – ₹ 42,750

= ₹ 5,973

Step V. Decision rule: Since NPV is positive we should accept the proposal to replace the machine.

SM Ch 3 · Illustration 13 (under 5.2 Optimum Replacement Cycle)

Illustration 13

X Ltd. is a taxi operator. Each taxi cost to company ₹ 4,00,000 and has a useful life of 3 years. The taxi's operating cost for each of 3 years and salvage value at the end of year is as follows:

	Year 1	Year 2	Year 3
Operating Cost	₹ 1,80,000	₹ 2,10,000	₹ 2,38,000
Resale Value	₹ 2,80,000	₹ 2,30,000	₹ 1,68,000

You are required to determine the optimal replacement period of taxi if cost of capital of X Ltd. is 10%.

Solution — ICAI

Solution

NPV if taxi is kept for 1 Year

= – ₹ 4,00,000 - ₹ 1,80,000 (0.909) + ₹ 2,80,000 (0.909)

= – ₹ 3,09,100

NPV if taxi is kept for 2 Year

= – ₹ 4,00,000 – ₹ 1,80,000 x 0.909 + ₹ 20,000 x 0.826

= – ₹ 5,47,100

NPV if taxi is kept for 3 Year

= – ₹ 4,00,000 – ₹ 1,80,000 x 0.909 – ₹ 2,10,000 x 0.826 – ₹ 70,000 x 0.751

= – ₹ 7,89,650

Since above NPV figures relate to different periods, there are not comparable. to make them comparable we shall use concept of EAC as follows:

EAC of 1 year

$$\frac{3,09,100}{0.909} = ₹\ 3,40,044$$

EAC of 2 year

$$\frac{5,47,100}{1.735} = ₹\ 3,15,331$$

EAC of 3 year

$$\frac{7,89,650}{2.486} = ₹\ 3,17,639$$

Since lowest EAC incur if taxi for 2 year; Hence the optimum replacement cycle to replace taxi in 2 years.

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. Skylark Airways is planning to acquire a light commercial aircraft for flying class clients at an investment of ₹ 50,00,000. The expected cash flow after tax for the next three years is as follows: (₹)

Year 1		Year 2		Year 3	
CFAT	Probability	CFAT	Probability	CFAT	Probability
14,00,000	0.1	15,00,000	0.1	18,00,000	0.2
18,00,000	0.2	20,00,000	0.3	25,00,000	0.5
25,00,000	0.4	32,00,000	0.4	35,00,000	0.2
40,00,000	0.3	45,00,000	0.2	48,00,000	0.1

The Company wishes to take into consideration all possible risk factors relating to airline operations. The company wants to know:

(i) The expected NPV of this venture assuming independent probability distribution with 6 per cent risk free rate of interest.

(ii) The possible deviation in the expected value.

(iii) How would standard deviation of the present value distribution help in Capital Budgeting decisions?

Solution — ICAI

1. (i) Expected NPV

(₹ in lakhs)

Year I			Year II			Year III		
CFAT	P	CF×P	CFAT	P	CF×P	CFAT	P	CF×P
14	0.1	1.4	15	0.1	1.5	18	0.2	3.6
18	0.2	3.6	20	0.3	6.0	25	0.5	12.5
25	0.4	10.0	32	0.4	12.8	35	0.2	7.0
40	0.3	12.0	45	0.2	9	48	0.1	4.8
	$\bar{x}$ or $\overline{CF}$	27.0		$\bar{x}$ or $\overline{CF}$	29.3		$\bar{x}$ or $\overline{CF}$	27.9

NPV	PV factor @ 6%	Total PV
27	0.943	25.461
29.3	0.890	26.077
27.9	0.840	23.436
	PV of cash inflow	74.974
	Less: Cash outflow	50.000
	NPV	24.974

(ii) Possible deviation in the expected value

Year I				
$X - \bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$	P_1	$(X - \bar{X})^2 P^1$
14 – 27	-13	169	0.1	16.9
18 – 27	-9	81	0.2	16.2

25 – 27	-2	4	0.4	1.6
40 – 27	13	169	0.3	50.7
				85.4

$\sigma_1 = \sqrt{85.4} = 9.241$

Year II				
$X - \bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$	P_2	$(X - \bar{X})^2 \times P_2$
15-29.3	-14.3	204.49	0.1	20.449
20-29.3	-9.3	86.49	0.3	25.947
32-29.3	2.7	7.29	0.4	2.916
45-29.3	15.7	246.49	0.2	49.298
				98.61

$\sigma_2 = \sqrt{98.61} = 9.930$

Year III				
$X - \bar{X}$	$X - \bar{X}$	$(X - \bar{X})^2$	P_3	$(X - \bar{X})^2 \times P_3$
18-27.9	-9.9	98.01	0.2	19.602
25-27.9	-2.9	8.41	0.5	4.205
35-27.9	7.1	50.41	0.2	10.082
48-27.9	20.1	404.01	0.1	40.401
				74.29

$\sigma_3 = \sqrt{74.29} = 8.619$

Standard deviation about the expected value:

$$\sigma = \sqrt{\frac{85.4}{(1.06)^2} + \frac{98.61}{(1.06)^4} + \frac{74.29}{(1.06)^6}} = 14.3696$$

(iii) Standard deviation is a statistical measure of dispersion; it measures the deviation from a central number i.e. the mean.

In the context of capital budgeting decisions especially where we take up two or more projects giving somewhat similar mean cash flows, by calculating standard deviation in such cases, we can measure in each case the extent of variation. It can then be used to identify which of the projects is least risky in terms of variability of cash flows.

A project, which has a lower coefficient of variation will be preferred if sizes are heterogeneous.

Besides this, if we assume that probability distribution is approximately normal we are able to calculate the probability of a capital budgeting project generating a net present value less than or more than a specified amount.

Practical Q2

2. Cyber Company is considering two mutually exclusive projects. Investment outlay of both the projects is ₹ 5,00,000 and each is expected to have a life of 5 years. Under three possible situations their annual cash flows and probabilities are as under:

		Cash Flow (₹)	
Situation	Probabilities	Project A	ProjectB
Good	0.3	6,00,000	5,00,000
Normal	0.4	4,00,000	4,00,000
Worse	0.3	2,00,000	3,00,000

The cost of capital is 7 per cent, which project should be accepted? Explain with workings.

Solution — ICAI

2. Project A

Expected Net Cash flow (ENCF)

0.3 (6,00,000) + 0.4 (4,00,000) + 0.3 (2,00,000) = 4,00,000

σ^2=0.3 (6,00,000– 4,00,000)^2 + 0.4 (4,00,000 – 4,00,000)^2 + 0.3 (2,00,000 – 4,00,000)^2

$\sigma = \sqrt{24,00,00,00,000}$

σ = 1,54,919.33

Present Value of Expected Cash Inflows = 4,00,000 × 4.100 = 16,40,000

NPV = 16,40,000 – 5,00,000 = 11,40,000

Project B

ENCF = 0.3 (5,00,000) + 0.4 (4,00,000) + 0.3 (3,00,000) = 4,00,000

σ^2=0.3 (5,00,000 – 4,00,000)^2 + 0.4 (4,00,000 – 4,00,000)^2 + 0.3 (3,00,000 – 4,00,000)^2

$\sigma = \sqrt{6,00,00,00,000}$

σ = 77,459.66

Present Value of Expected Cash Inflows = 4,00,000 × 4.100 = 16,40,000

NPV = 16,40,000 – 5,00,000 = 11,40,000

Recommendation: NPV in both projects being the same, the project should be decided on the basis of standard deviation and hence project 'B' should be accepted having lower standard deviation, means less risky.

Practical Q3

3. A company is considering Projects X and Y with following information:

Project	Expected NPV (₹)	Standard deviation
X	1,22,000	90,000
Y	2,25,000	1,20,000

(i) Which project will you recommend based on the above data?

(ii) Explain whether your opinion will change, if you use coefficient of variation as a measure of risk.

(iii) Which measure is more appropriate in this situation and why?

Solution — ICAI

3. (i) On the basis of standard deviation project X be chosen because it is less risky than Project Y having higher standard deviation.

(ii) $CV_x = \frac{SD}{ENPV} = \frac{90,000}{1,22,000} = 0.738$

$CV_y = \frac{1,20,000}{2,25,000} = 0.533$

On the basis of Co-efficient of Variation (C.V.) Project X appears to be riskier and hence Y should be accepted.

(iii) However, the NPV method in such conflicting situation is best because the NPV method is in compatibility of the objective of wealth maximisation in terms of time value.

Practical Q4

4. KLM Ltd., is considering taking up one of the two projects-Project-K and Project-So Both the projects having same life require equal investment of ₹ 80 lakhs each. Both are estimated to have almost the same yield. As the company is new to this type of business, the cash flow arising from the projects cannot be estimated with certainty. An attempt was therefore, made to use probability to analyse the pattern of cash flow from other projects during the first year of operations. This pattern is likely to continue during the life of these projects. The results of the analysis are as follows:

Project K		Project S	
Cash Flow (in ₹)	Probability	Cash Flow (in ₹)	Probability
11	0.10	09	0.10
13	0.20	13	0.25
15	0.40	17	0.30
17	0.20	21	0.25
19	0.10	25	0.10

Required:

(i) Calculate variance, standard deviation and co-efficient of variance for both the projects.

(ii) Which of the two projects is riskier?

Solution — ICAI

4. Calculation of Variance and Standard Deviation

Project K

Expected Net Cash Flow

= (0.10 x 11) + (0.20 x13) + (0.40 x 15) + (0.20 x 17) + (0.10 x 19)

= 1.1 + 2.6 + 6 + 3.4 + 1.9 = 15

$$\sigma^2 = 0.10(11-15)^2 + 0.20(13-15)^2 + 0.40(15-15)^2 + 0.20(17-15)^2 + 0.10(19-15)^2$$

= 1.6 + 0.8 + 0 + 0.8 + 1.6 = 4.8

$$\sigma = \sqrt{4.8} = 2.19$$

Project S

Expected Net Cash Flow

= (0.10 X 9) + (0.25 X 13) + (0.30 X 17) + (0.25 X 21) + (0.10 X 25)

= 0.9 + 3.25 + 5.1 + 5.25 + 2.5 = 17

$$\sigma^2 = 0.1(9-17)^2 + 0.25(13-17)^2 + 0.30(17-17)^2 + 0.25(21-17)^2 + 0.10(25-17)^2$$

= 6.4 + 4 + 0 + 4 + 6.4 = 20.8

$$\sigma = \sqrt{20.8} = 4.56$$

Calculation of Coefficient of Variation

$$\text{Coefficient of Variation} = \frac{\text{Standard Deviation}}{\text{Mean}}$$

$$\text{Project K} = \frac{2.19}{15} = 0.146$$

$$\text{Project S} = \frac{4.56}{17} = 0.268$$

Project S is riskier as it has higher Coefficient of Variation.

Practical QS

5. Project X and Project Y are under the evaluation of XY Co. The estimated cash flows and their probabilities are as below:

Project X : Investment (year 0) ₹ 70 lakhs

Probability weights	0.30	0.40	0.30
Years	₹ lakhs	₹ lakhs	₹ lakhs
1	30	50	65
2	30	40	55
3	30	40	45

Project Y: Investment (year 0) ₹ 80 lakhs.

Probability weighted	Annual cash flows through life
	₹ lakhs
0.20	40
0.50	45
0.30	50

(a) Which project is better based on NPV, criterion with a discount rate of 10%?

(b) Compute the standard deviation of the present value distribution and analyse the inherent risk of the projects.

Solution — ICAI

5. (a) Calculation of NPV of XY Co.:

Project X		Cash flow	PVF	PV
Year				
1	(30 × 0.3) + (50 × 0.4) + (65 × 0.3)	48.5	0.909	44.09
2	(30 × 0.3) + (40 × 0.4) + (55 × 0.3)	41.5	0.826	34.28
3	(30 × 0.3) + (40 × 0.4) + (45 × 0.3)	38.5	0.751	28.91
				107.28
	NPV: (107.28 – 70.00) =			(+) 37.28

Project Y (For 1-3 Years)

1-3	(40 × 0.2) + (45 × 0.5) + (50 × 0.3)	45.5	2.487	113.16
	NPV (113.16 – 80.00)			(+) 33.16

(b) Calculation of Standard deviation σ

As per Hiller's model

$$M = \sum_{i=0}^{n} (1+r)^{-1} Mi$$

$$\sigma^2 = \sum_{i=0}^{n} (1+r)^{-2i} \sigma_i^2$$

Hence

Project X

Year

1 $\sqrt{(30-48.5)^2 0.30 + (50-48.5)^2 0.40 + (65-48.5)^2 0.30} = \sqrt{185.25} = 13.61$

2 $\sqrt{(30-41.5)^2 0.30 + (40-41.5)^2 0.40 + (55-41.5)^2 0.30} = \sqrt{95.25} = 9.76$

3 $\sqrt{(30-38.5)^2 0.30 + (40-38.5)^2 0.40 + (45-38.5)^2 0.30} = \sqrt{35.25} = 5.94$

Standard Deviation about the expected value

$$= \sqrt{\frac{185.25}{(1+0.10)^2} + \frac{95.25}{(1+0.10)^4} + \frac{35.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{185.25}{1.21} + \frac{95.25}{1.4641} + \frac{35.25}{1.7716}} = \sqrt{153.10+65.06+19.90}$$

$$= \sqrt{238.06} = 15.43$$

Project Y (For 1-3 Years)

$$\sqrt{(40-45.5)^2 0.20 + (45-45.5)^2 0.50 + (50-45.5)^2 0.30} \quad = \sqrt{12.25} \quad = 3.50$$

Standard Deviation about the expected value

$$= \sqrt{\frac{12.25}{(1+0.10)^2} + \frac{12.25}{(1+0.10)^4} + \frac{12.25}{(1+0.10)^6}}$$

$$= \sqrt{\frac{12.25}{1.21} + \frac{12.25}{1.4641} + \frac{12.25}{1.7716}} = \sqrt{10.12+8.37+6.91}$$

$$= \sqrt{25.4} = 5.03$$

Analysis: Project Y is less risky as its Standard Deviation is less than Project X.

Practical Q6

6. Shivam Ltd. is considering two mutually exclusive projects A and B. Project A costs ₹ 36,000 and project B ₹ 30,000. You have been given below the net present value probability distribution for each project.

Project A		Project B	
NPV estimates (₹)	Probability	NPV estimates (₹)	Probability
15,000	0.2	15,000	0.1
12,000	0.3	12,000	0.4
6,000	0.3	6,000	0.4
3,000	0.2	3,000	0.1

(i) Compute the expected net present values of projects A and B.

(ii) Compute the risk attached to each project i.e. standard deviation of each probability distribution.

(iii) Compute the profitability index of each project.

(iv) Which project do you recommend? State with reasons.

Solution — ICAI

6. (i) Statement showing computation of expected net present value of Projects A and B:

Project A			Project B		
NPV Estimate (₹)	Probability	Expected Value	NPV Estimate	Probability	Expected Value
15,000	0.2	3,000	15,000	0.1	1,500
12,000	0.3	3,600	12,000	0.4	4,800
6,000	0.3	1,800	6,000	0.4	2,400
3,000	0.2	600	3,000	0.1	300
	1.0	EV = 9,000		1.0	EV = 9,000

(ii) Computation of Standard deviation of each project

Project A

P	X	(X – EV)	P (X - EV)²
0.2	15,000	6,000	72,00,000
0.3	12,000	3,000	27,00,000
0.3	6,000	- 3,000	27,00,000
0.2	3,000	- 6,000	72,00,000
			Variance = 1,98,00,000

Standard Deviation of Project A = $\sqrt{1,98,00,000}$ = ₹ 4,450

Project B

P	X	(X – EV)	P (X - EV)²
0.1	15,000	6,000	36,00,000
0.4	12,000	3,000	36,00,000
0.4	6,000	- 3,000	36,00,000
0.1	3,000	- 6,000	36,00,000
			Variance = 1,44,00,000

Standard Deviation of Project A = $\sqrt{1,44,00,000}$ = ₹ 3,795

(iii) Computation of profitability of each project

Profitability index = Discount cash inflow / Initial outlay

In case of Project A : PI = $\frac{9,000+36,000}{36,000}=\frac{45,000}{36,000}=1.25$

In case of Project B : PI = $\frac{9,000+30,000}{30,000}=\frac{39,000}{30,000}=1.30$

(iv) Measurement of risk is made by the possible variation of outcomes around the expected value and the decision will be taken in view of the variation in the expected value where two projects have the same expected value, the decision will be the project which has smaller variation in expected value. In the selection of one of the two projects A and B, Project B is preferable because the possible profit which may occur is subject to less variation (or dispersion). Much higher risk is lying with project A.

Practical Q7

7. Following are the estimates of the net cash flows and probability of a new project of M/s X Ltd.:

	Year	P = 0.3	P = 0.5	P = 0.2
Initial investment	0	4,00,000	4,00,000	4,00,000
Estimated net after tax cash inflows per year	1 to 5	1,00,000	1,10,000	1,20,000
Estimated salvage value (after tax)	5	20,000	50,000	60,000

Required rate of return from the project is 10%. Find:

(i) The expected NPV of the project.

(ii) The best case and the worst case NPVs.

(iii) The probability of occurrence of the worst case if the cash flows are perfectly dependent overtime and independent overtime.

(iv) Standard deviation and coefficient of variation assuming that there are only three streams of cash flow, which are represented by each column of the table with the given probabilities.

(v) Coefficient of variation of X Ltd. on its average project which is in the range of 0.95 to 1.0. If the coefficient of variation of the project is found to be less risky than average, 100 basis points are deducted from the Company's cost of Capital

Should the project be accepted by X Ltd?

Solution — ICAI

7. (a) (i) Expected cash flows:-

Year			Net cash flows	P.V.	PV. @ 10%
0	(4,00,000 x 1)	=	(-)4,00,000	1.000	(-)4,00,000
1 to 4	(1,00,000x0.3+1,10,000x0.5 + 1,20,000 x 0.2)	=	1,09,000	3.170	3,45,530
5	[1,09,000 + (20,000 x 0.3 + 50,000 x 0.5 + 60,000 x 0.2)]	=	1,52,000	0.621	94,392
	NPV =				39,922

(ii) ENPV of the worst case

1,00,000 x 3.790 = ₹ 3,79,000 (Students may have 3.791 also the values will change accordingly)

20,000 x 0.621 = ₹ 12,420/-

ENPV = (-) 4,00,000 + 3,79,000 + 12,420 = (-) ₹ 8,580/-

ENPV of the best case

ENPV = (-) 4,00,000 + 1,20,000 x 3.790 + 60,000 x 0.621 = ₹ 92,060/-.

(iii) (a) Required probability = 0.3

(b) Required probability = $(0.3)^5 = 0.00243$

(iv) The base case NPV = (-) 4,00,000 + (1,10,000 x 3.79) + (50,000 x 0.621)

= ₹ 47,950/-

ENPV = 0.30 x (-) 8580 + 0.5 x 47950 + 92060 x 0.20 = ₹ 39,813/-

Therefore,

$$\sigma_{ENPV} = \sqrt{0.3(-8580-39{,}813)^2 + 0.5(47950-39813)^2 + 0.2(92{,}060-39{,}813)^2}$$

= ₹ 35,800/-

Therefore, CV = 35,800/39,813 = 0.90

(v) Risk adjusted out of cost of capital of X Ltd. = 10% - 1% = 9%.

NPV

Year	Expected net cash flow	PV @ 9%	
0	(-) 4,00,000	1.000	(-) 4,00,000
1 to 4	1,09,000	3.240	3,53,160
5	1,52,000	0.650	98,800
		ENPV =	51,960

Therefore, the project should be accepted.

Practical Q8

8. XY Ltd. has under its consideration a project with an initial investment of ₹ 1,00,000. Three probable cash inflow scenarios with their probabilities of occurrence have been estimated as below:

Annual cash inflow (₹)	20,000	30,000	40,000
Probability	0.1	0.7	0.2

The project life is 5 years and the desired rate of return is 20%. The estimated terminal values for the project assets under the three probability alternatives, respectively, are ₹ 0, 20,000 and 30,000.

You are required to:

(i) Find the probable NPV;

(ii) Find the worst-case NPV and the best-case NPV; and

(iii) State the probability occurrence of the worst case, if the cash flows are perfectly positively correlated over time.

Solution — ICAI

8. The expected cash flows of the project are as follows:

Year	Pr = 0.1	Pr = 0.7	Pr = 0.2	Total
	₹	₹	₹	₹
0	-10,000	-70,000	-20,000	-1,00,000
1	2,000	21,000	8,000	31,000
2	2,000	21,000	8,000	31,000
3	2,000	21,000	8,000	31,000
4	2,000	21,000	8,000	31,000
5	2,000	21,000	8,000	31,000
5	0	14,000	6,000	20,000

(i) NPV based on expected cash flows would be as follows:

$$=-₹\ 1{,}00{,}000 + \frac{₹\ 31{,}000}{(1+0.20)^1} + \frac{₹\ 31{,}000}{(1+0.20)^2} + \frac{₹\ 31{,}000}{(1+0.20)^3} + \frac{₹\ 31{,}000}{(1+0.20)^4} + \frac{₹\ 31{,}000}{(1+0.20)^5} + \frac{₹\ 20{,}000}{(1+0.20)^5}$$

= - ₹ 1,00,000 + ₹ 25,833.33 + ₹ 21,527.78 + ₹ 17,939.81 + ₹ 14,949.85+ ₹ 12,458.20 + ₹ 8,037.55

NPV = ₹ 746.52

(ii) For the worst case, the cash flows from the cash flow column farthest on the left are used to calculate NPV

$$= - ₹\ 100{,}000 + \frac{₹ 20{,}000}{(1+0.20)^1} + \frac{₹ 20{,}000}{(1+0.20)^2} + \frac{₹ 20{,}000}{(1+0.20)^3} + \frac{₹ 20{,}000}{(1+0.20)^4} + \frac{₹ 20{,}000}{(1+0.20)^5}$$

= - ₹ 100,000 + ₹ 16,666.67+ ₹ 13,888.89 + ₹ 11,574.07 + ₹ 9,645.06+ ₹ 8037.76

NPV = - ₹ 40,187.76

For the best case, the cash flows from the cash flow column farthest on the right are used to calculated NPV

$$= -₹\ 100{,}000 + \frac{₹\ 40{,}000}{(1+0.20)^1} + \frac{₹\ 40{,}000}{(1+0.20)^2} + \frac{₹\ 40{,}000}{(1+0.20)^3} + \frac{₹\ 40{,}000}{(1+0.20)^4} + \frac{₹\ 40{,}000}{(1+0.20)^5} + \frac{₹\ 30{,}000}{(1+0.20)^5}$$

= - ₹ 1,00,000 + ₹ 33,333.33+ ₹ 27,777.78 + ₹ 23,148.15+ ₹ 19,290.12 + ₹ 16,075.10 + ₹ 12,056.33

NPV = ₹ 31,680.81

(iii) If the cash flows are perfectly dependent, then the low cash flow in the first year will mean a low cash flow in every year. Thus, the possibility of the worst case occurring is the probability of getting ₹ 20,000 net cash flow in year 1 is 10%.

Practical Q9

9. XYZ Ltd. is considering a project for which the following estimates are available:

	₹
Initial Cost of the project	10,00,000
Sales price/unit	60
Cost/unit	40
Sales volumes	
Year 1	20000 units
Year 2	30000 units
Year 3	30000 units

Discount rate is 10% p.a.

You are required to measure the sensitivity of the project in relation to each of the following parameters:

(a) Sales Price/unit

(b) Unit cost

(c) Sales volume

(d) Initial outlay and

(e) Project lifetime

Taxation may be ignored.

Solution — ICAI

9. Calculation of NPV

$$NPV = -10,00,000 + \frac{20,000\times20}{1.1} + \frac{30,000\times20}{1.21} + \frac{30,000\times20}{1.331}$$

= - ₹ 10,00,000 + ₹ 3,63,636 + ₹ 4,95,868 + ₹ 4,50,789

= ₹ 13,10,293 – ₹ 10,00,000

= ₹ 3,10,293

Measurement of sensitivity is as follows:

(a) Sales Price:-

Let the sale price/Unit be S so that the project would break even with 0 NPV.

$$\therefore 10,00,000 = \frac{20,000\times(S-40)}{1.1} + \frac{30,000\times(S-40)}{1.21} + \frac{30,000(S-40)}{1.331}$$

S – 40 = ₹ 10,00,000/₹ 65,514

S – 40 = ₹ 15.26

S = ₹ 55.26 which represents a fall of (60-55.26)/60

Or 0.079 or 7.9%

Alternative Method

$$\frac{10,00,000 \times 20}{13,10,293} = ₹\ 15.26$$

S= ₹ 40 + ₹ 15.26

= ₹ 55.26

Alternative Solution

If sale Price decreased by say 10%, then NPV (at Sale Price of ₹ 60 – ₹ 6 = ₹ 54)

$$NPV = -10,00,000 + \frac{20000 \times 14}{(1.1)^1} + \frac{30000 \times 14}{(1.1)^2} + \frac{30000 \times 14}{(1.1)^3}$$

= -₹ 10,00,000 + ₹ 2,54,545 + ₹ 3,47,107 + ₹ 3,15,552

= - ₹ 82,796

$$\text{NPV decrease (\%)} = \frac{3,10,293-(-82,796)}{3,10,293} \times 100 = 126.68\%$$

(b) Unit Cost:-

If sales price = ₹ 60 the cost price required to give a margin of ₹ 15.26 is (₹ 60 – ₹ 15.26) or ₹ 44.74 which would represent a rise of 11.85% i.e., $\left(\frac{44.74-40}{40} \times 100\right)$

Alternative Solution

If unit cost increased by say 10%. The new NPV will be as follows:

$$NPV = -10,00,000 + \frac{20000 \times 16}{(1.1)^1} + \frac{30000 \times 16}{(1.1)^2} + \frac{30000 \times 16}{(1.1)^3}$$

= -₹ 10,00,000 + ₹ 2,90,909 + ₹ 3,96,694 + ₹ 3,60,631

= ₹ 48,234

$$\text{NPV decrease (\%)} = \frac{3,10,293-(48,234)}{3,10,293} \times 100 = 84.46\%$$

(c) Sales volume:-

The requisite percentage fall is:-

3,10,293/13,10,293 × 100 = 23.68%

Alternative Solution

If sale volume decreased by say 10%. The new NPV will be as follows:

$$NPV = -10,00,000 + \frac{18000\times20}{(1.1)^1} + \frac{27000\times20}{(1.1)^2} + \frac{27000\times20}{(1.1)^3}$$

= - ₹ 10,00,000 + ₹ 3,27,272 + ₹ 4,46,281 + ₹ 4,05,710

= ₹ 1,79,263

$$\text{NPV decrease (\%)} = \frac{3,10,293 - 1,79,263}{3,10,293} \text{ X } 100 = 42.22\%$$

(d) Since PV of inflows remains at ₹13,10,293 the initial outlay must also be the same.

∴ Percentage rise = 3,10,293/10,00,000 × 100 = 31.03%.

Alternative Solution

If initial outlay increased by say 10%. The new NPV will be as follows:

$$NPV = -₹\ 11,00,000 + \frac{20000\times20}{(1.1)^1} + \frac{30000\times20}{(1.1)^2} + \frac{30000\times20}{(1.1)^3}$$

= - ₹ 11,00,000 + ₹ 3,63,636 + ₹ 4,95,868 + ₹ 4,50,789 = ₹ 2,10,293

$$\text{NPV decrease (\%)} = \frac{3,10,293 - 2,10,293}{3,10,293} \text{ X } 100 = 32.22\%$$

(e) Present value for 1st two years.

= - ₹ 10,00,000 + ₹ 4,00,000 x 0.909 + ₹ 6,00,000 x 0.826

= - ₹ 10,00,000 + ₹ 3,63,600 + ₹ 4,95,600

= - ₹ 10,00,000 + ₹ 8,59,200

= - ₹ 1,40,800

∴ The project needs to run for some part of the third year so that the present value of return is ₹ 1,40,800. It can be computed as follows:

(i) 30,000 units x ₹ 20 x 0.751 = ₹ 4,50,600

(ii) Per day Production in (₹) assuming a year of 360 days =

$$\frac{₹\ 4,50,600}{360} = ₹\ 1,252$$

(iii) Days needed to recover ₹1,40,800 = $\frac{₹\ 1,40,800}{₹\ 1,252} = 112$

Thus, if the project runs for 2 years and 112 days then break even would be achieved representing a fall of $\frac{(3 - 2.311)}{3} \times 100 = 22.97\%$.

Practical Q10

10. From the following details relating to a project, analyse the sensitivity of the project to changes in initial project cost, annual cash inflow and cost of capital:

Initial Project Cost (₹)	1,20,000
Annual Cash Inflow (₹)	45,000
Project Life (Years)	4
Cost of Capital	10%

To which of the three factors, the project is most sensitive? (Use annuity factors: for 10% 3.169 and 11% 3.103).

Solution — ICAI

10. CALCULATION OF NPV

	₹
PV of cash inflows (₹ 45,000 x 3.169)	1,42,605
Initial Project Cost	1,20,000
NPV	22,605
If initial project cost is varied adversely by 10%*	
NPV (Revised) (₹ 1,42,605 - ₹ 1,32,000)	₹ 10,605
Change in NPV (₹ 22,605 – ₹ 10,605)/ ₹ 22,605 i.e.	53.08 %
If annual cash inflow is varied adversely by 10%*	
Revised annual inflow	₹ 40,500
NPV (Revised) (₹ 40,500 x 3.169) – (₹ 1,20,000)	(+) ₹ 8,345
Change in NPV (₹ 22,605 – ₹ 8,345) / ₹ 22,605	63.08 %
If cost of capital is varied adversely by 10%*	
NPV (Revised) (₹ 45,000 x 3.103) – ₹ 1,20,000	(+) ₹ 19,635
Change in NPV (₹ 22,605 – ₹ 19,635) / ₹ 22,605	13.14 %

Conclusion: Project is most sensitive to 'annual cash inflow'.

*Note: Students may please note that they may assume any other percentage rate other than 10 % say 15%, 20 % 25 % etc.

Practical Q11

11. Red Ltd. is considering a project with the following Cash flows:

₹

Years	Cost of Plant	Recurring Cost	Savings
0	10,000		
1		4,000	12,000
2		5,000	14,000

The cost of capital is 9%. Measure the sensitivity of the project to changes in the levels of plant value, running cost and savings (considering each factor at a time) such that the NPV becomes zero. The P.V. factor at 9% are as under:

Year	Factor
0	1
1	0.917
2	0.842

Which factor is the most sensitive to affect the acceptability of the project?

Solution — ICAI

11. P.V. of Cash Flows

Year 1	Running Cost	₹ 4,000 x 0.917	= (₹ 3,668)
	Savings	₹ 12,000 x 0.917	= ₹ 11,004
Year 2	Running Cost	₹ 5,000 x 0.842	= (₹ 4,210)
	Savings	₹ 14,000 x 0.842	= ₹ 11,788
			₹ 14,914
Year 0	*Less:* P.V. of Cash Outflow	₹ 10,000 x 1	₹ 10,000
		NPV	₹ 4,914

Sensitivity Analysis

(i) Increase of Plant Value by ₹ 4,914

$$\therefore \frac{4,914}{10,000} \times 100 = 49.14\%$$

(ii) Increase in PV of Running Cost by ₹ 4,914

$$\frac{4,914}{3,668+4,210} = \frac{4,914}{7,878} \times 100 = 62.38\%$$

(iii) Fall in PV of Saving by ₹ 4,914

$$\frac{4,914}{11,004+11,788} = \frac{4,914}{22,792} \times 100 = 21.56\%$$

Hence, savings factor is the most sensitive to affect the acceptability of the project as in comparison of other two factors a slight % change in this fact shall more affect the NPV than others.

Alternative Solution

P.V. of Cash Flows

Year 1	Running Cost	₹ 4,000 x 0.917	= (₹ 3,668)
	Savings	₹ 12,000 x 0.917	= ₹ 11,004
Year 2	Running Cost	₹ 5,000 x 0.842	= (₹ 4,210)
	Savings	₹ 14,000 x 0.842	= ₹ 11,788
			₹ 14,914
Year 0	Less: P.V. of Cash Outflow	₹ 10,000 x 1	₹ 10,000
		NPV	₹ 4,914

Sensitivity Analysis

(i) If the initial project cost is varied adversely by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 1,000) = ₹ 3,914

$$\text{Change in NPV } \frac{₹\ 4,914 - ₹\ 3,914}{₹\ 4,914} = 20.35\%$$

(ii) If Annual Running Cost is varied by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 400 X 0.917 – ₹ 500 X 0.842)

= ₹ 4,914 – ₹ 367 – ₹ 421= ₹ 4,126

Change in NPV $\frac{₹\ 4,914 - ₹\ 4,126}{₹\ 4,914}$ = 16.04%

(iii) If Saving is varied by say 10%*.

NPV (Revised) (₹ 4,914 – ₹ 1,200 X 0.917 – ₹ 1,400 X 0.842)

= ₹ 4,914 – ₹ 1,100 – ₹ 1,179 = ₹ 2,635

Change in NPV $\frac{4,914 - 2,635}{4,914}$ = 46.38%

Hence, savings factor is the most sensitive to affect the acceptability of the project.

* Any percentage of variation other than 10% can also be assumed.

Practical Q12

12. The Easygoing Company Limited is considering a new project with initial investment, for a product "Survival". It is estimated that IRR of the project is 16% having an estimated life of 5 years.

Financial Manager has studied that project with sensitivity analysis and informed that annual fixed cost sensitivity is 7.8416%, whereas cost of capital (discount rate) sensitivity is 60%.

Other information available are:

Profit Volume Ratio (P/V) is 70%,

Variable cost ₹ 60/- per unit

Annual Cash Flow ₹ 57,500/-

Ignore Depreciation on initial investment and impact of taxation.

Calculate

(i) Initial Investment of the Project

(ii) Net Present Value of the Project

(iii) Annual Fixed Cost

(iv) Estimated annual unit of sales

(v) Break Even Units

Cumulative Discounting Factor for 5 years

8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%
3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127

Solution — ICAI

12. (i) Initial Investment

IRR = 16% (Given)

At IRR, NPV shall be zero, therefore

Initial Cost of Investment = PVAF (16%,5) x Cash Flow (Annual)

= 3.274 x ₹ 57,500

= ₹ 1,88,255

(ii) Net Present Value (NPV)

Let Cost of Capital be X, then $\frac{16-X}{X} = 60\%$ X = 10%

Thus NPV of the project

= Annual Cash Flow x PVAF (10%, 5) – Initial Investment

= ₹ 57,500 x 3.791 – ₹ 1,88,255

= ₹ 2,17,982.50 – ₹ 1,88,255 = ₹ 29,727.50

(iii) Annual Fixed Cost

Let change in the Fixed Cost which makes NPV zero is X. Then,

₹ 29,727.50 – 3.791X = 0

Thus X = ₹ 7,841.60

Let original Fixed Cost be Y then,

Y × 7.8416% = ₹ 7,841.60

Y = ₹ 1,00,000

Thus Fixed Cost is equal to ₹ 1,00,000

(iv) Estimated Annual Units of Sales

$$\text{Selling Price per unit} = \frac{₹\ 60}{100\% - 70\%} = ₹\ 200$$

$$\frac{\text{Annual Cash Flow + Fixed Cost}}{\text{P/V Ratio}} = \text{Sales Value}$$

$$\frac{₹\ 57,500 + ₹\ 1,00,000}{0.70} = ₹\ 2,25,000$$

$$\text{Sales in Units} = \frac{₹\ 2,25,000}{₹\ 200} = 1,125 \text{ units}$$

(v) Break Even Units

$$\frac{\text{Fixed Cost}}{\text{Contribution Per Unit}} = \frac{1,00,000}{140} = 714.285 \text{ units}$$

Practical Q13

13. Unnat Ltd. is considering investing ₹ 50,00,000 in a new machine. The expected life of machine is five years and has no scrap value. It is expected that 2,00,000 units will be produced and sold each year at a selling price of ₹ 30.00 per unit. It is expected that the variable costs to be ₹ 16.50 per unit and fixed costs to be ₹ 10,00,000 per year. The cost of capital of Unnat Ltd. is 12% and acceptable level of risk is 20%.

You are required to measure the sensitivity of the project's net present value to a change in the following project variables:

(a) sale price;

(b) sales volume;

(c) variable cost;

(d) On further investigation it is found that there is a significant chance that the expected sales volume of 2,00,000 units per year will not be achieved. The sales manager of Unnat Ltd. suggests that sales volumes could depend on expected economic states which could be assigned the following probabilities:

State of Economy	Annual Sales (in Units)	Prob.
Poor	1,75000	0·30
Normal	2,00,000	0·60
Good	2,25,000	0·10

Calculate expected net present value of the project and give your decision whether company should accept the project or not.

Solution — ICAI

13. Calculation of NPV

= - ₹ 50,00,000 + [2,00,000 (₹ 30 – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

= - ₹ 50,00,000 + [2,00,000 (₹ 13.50) – ₹ 10,00,000] 3.605

= - ₹ 50,00,000 + [₹ 27,00,000 – ₹ 10,00,000] 3.605

= - ₹ 50,00,000 + ₹ 61,28,500 = ₹ 11,28,500

Measurement of Sensitivity Analysis

(a) Sales Price:-

Let the sale price/Unit be S so that the project would break even with 0 NPV.

∴ ₹ 50,00,000 = [2,00,000 (S – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [2,00,000S – ₹ 33,00,000 – ₹ 10,00,000] 3.605

₹ 50,00,000 = [2,00,000S – ₹ 43,00,000] 3.605

₹ 13,86,963 = 2,00,000S – ₹ 43,00,000

₹ 56,86,963 = 2,00,000S

S = ₹ 28.43 which represents a fall of (30 - 28.43)/30 or 0.0523 or 5.23%

(b) Sales volume:-

Let V be the sale volume so that the project would break even with 0 NPV.

₹ 50,00,000 = [V (₹ 30 – ₹ 16.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [V (₹ 13.50) – ₹ 10,00,000] PVIAF (12%,5)

₹ 50,00,000 = [₹ 13.50V – ₹ 10,00,000] 3.605

₹ 13,86,963 = ₹ 13.50V – ₹ 10,00,000

₹ 23,86,963 = ₹ 13.50V

V = 1,76,812 which represents a fall of (2,00,000 - 1,76,812)/2,00,000 or 0.1159 or 11.59%

(c) Variable Cost:

Let the variable cost be V so that the project would break even with 0 NPV.

∴ ₹ 50,00,000 = [2,00,000(₹ 30 – V) – ₹ 10,00,000] PVIAF(12%,5)

₹ 50,00,000 = [₹ 60,00,000 – 2,00,000 V – ₹ 10,00,000] 3.605

₹ 50,00,000 = [₹ 50,00,000 – 2,00,000 V] 3.605

₹ 13,86,963 = ₹ 50,00,000 – 2,00,000 V

₹ 36,13,037 = 2,00,000V

V = ₹ 18.07 which represents a fall of (18.07 – 16.50)/16.50 or 0.0951 or 9.51%

(d) Expected Net Present Value

(1,75,000 X 0.30) + (2,00,000 X 0.60) + (2,25,000 X 0.10) =1,95,000

NPV = [1,95,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = ₹ 8,85,163

Further NPV in worst and best cases will be as follows:

Worst Case:

[1,75,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = - ₹ 88,188

Best Case:

[2,25,000 X ₹ 13.50 – ₹ 10,00,000] 3.605 – ₹ 50,00,000 = ₹ 23,45,188

Thus, there are 30% chances that the rise will be a negative NPV and 70% chances of positive NPV. Since acceptable level of risk of Unnat Ltd. is 20% and there are 30% chances of negative NPV hence project should not be accepted.

Practical Q14

14. The Textile Manufacturing Company Ltd., is considering one of two mutually exclusive proposals, Projects M and N, which require cash outlays of ₹ 8,50,000 and ₹ 8,25,000 respectively. The certainty-equivalent (C.E) approach is used in incorporating risk in capital budgeting decisions. The current yield on government bonds is 6% and this is used as the risk free rate. The expected net cash flows and their certainty equivalents are as follows:

Project M			Project N	
Year-end	Cash Flow ₹	C.E.	Cash Flow ₹	C.E.
1	4,50,000	0.8	4,50,000	0.9
2	5,00,000	0.7	4,50,000	0.8
3	5,00,000	0.5	5,00,000	0.7

Present value factors of ₹ 1 discounted at 6% at the end of year 1, 2 and 3 are 0.943, 0.890 and 0.840 respectively.

Required:

(i) Which project should be accepted?

(ii) If risk adjusted discount rate method is used, which project would be appraised with a higher rate and why?

Solution — ICAI

14. (i) Statement Showing the Net Present Value of Project M

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor at 6% (d)	Total Present value (₹) (e) = (c) × (d)
1	4,50,000	0.8	3,60,000	0.943	3,39,480
2	5,00,000	0.7	3,50,000	0.890	3,11,500
3	5,00,000	0.5	2,50,000	0.840	2,10,000
					8,60,980
Less: Initial Investment					8,50,000
Net Present Value					10,980

Statement Showing the Net Present Value of Project N

Year end	Cash Flow (₹) (a)	C.E. (b)	Adjusted Cash flow (₹) (c) = (a) × (b)	Present value factor (d)	Total Present value (₹) (e) = (c) × (d)
1	4,50,000	0.9	4,05,000	0.943	3,81,915
2	4,50,000	0.8	3,60,000	0.890	3,20,400
3	5,00,000	0.7	3,50,000	0.840	2,94,000
					9,96,315
Less: Initial Investment					8,25,000
Net Present Value					1,71,315

Decision: Since the net present value of Project N is higher, so the project N should be accepted.

(ii) Certainty - Equivalent (C.E.) Co-efficient of Project M (2.0) is lower than Project N (2.4). This means Project M is riskier than Project N as "higher the riskiness of a cash flow, the lower will be the CE factor". If risk adjusted discount rate (RADR) method is used, Project M would be analysed with a higher rate.

RADR is based on the premise that riskiness of a proposal may be taken care of, by adjusting the discount rate. The cash flows from a more risky proposal should be discounted at a relatively higher discount rate as compared to other proposals whose cash flows are less risky. Any investor is basically risk averse. However, he may be ready to take risk provided he is rewarded for undertaking risk by higher returns. So, more risky the investment is, the greater would be the expected return. The expected return is expressed in terms of discount rate which is also the minimum required rate of return generated by a proposal if it is to be accepted. Therefore, there is a positive correlation between risk of a proposal and the discount rate.

Practical Q15

15. Determine the risk adjusted net present value of the following projects:

	X	Y	Z
Net cash outlays (₹)	2,10,000	1,20,000	1,00,000
Project life	5 years	5 years	5 years
Annual Cash inflow (₹)	70,000	42,000	30,000
Coefficient of variation	1.2	0.8	0.4

The Company selects the risk-adjusted rate of discount on the basis of the coefficient of variation:

Coefficient of Variation	Risk-Adjusted Rate of Return	P.V. Factor 1 to 5 years At risk adjusted rate of discount
0.0	10%	3.791
0.4	12%	3.605
0.8	14%	3.433
1.2	16%	3.274
1.6	18%	3.127
2.0	22%	2.864
More than 2.0	25%	2.689

Solution — ICAI

15. **Statement showing the determination of the risk adjusted net present value**

Projects	*Net cash outlays*	*Coefficient of variation*	*Risk adjusted discount rate*	*Annual cash inflow*	*PV factor 1-5 years*	*Discounted cash inflow*	*Net present value*
	₹			₹		₹	₹
(i)	*(ii)*	*(iii)*	*(iv)*	*(v)*	*(vi)*	*(vii) = (v) × (vi)*	*(viii) = (vii) – (ii)*
X	2,10,000	1.20	16%	70,000	3.274	2,29,180	19,180
Y	1,20,000	0.80	14%	42,000	3.433	1,44,186	24,186
Z	1,00,000	0.40	12%	30,000	3.605	1,08,150	8,150

Practical Q16

16. New Projects Ltd. is evaluating 3 projects, P-I, P-II, P-III. Following information is available in respect of these projects:

	P-I	P-II	P-III
Cost	₹ 15,00,000	₹ 11,00,000	₹ 19,00,000
Inflows-Year 1	6,00,000	6,00,000	4,00,000
Year 2	6,00,000	4,00,000	6,00,000
Year 3	6,00,000	5,00,000	8,00,000
Year 4	6,00,000	2,00,000	12,00,000
Risk Index	1.80	1.00	0.60

Minimum required rate of return of the firm is 15% and applicable tax rate is 40%. The risk free interest rate is 10%.

Required:

(i) Find out the risk-adjusted discount rate (RADR) for these projects.

(ii) Which project is the best?

Solution — ICAI

16. (i) The risk free rate of interest and risk factor for each of the projects are given. The risk adjusted discount rate (RADR) for different projects can be found on the basis of CAPM as follows:

Required Rate of Return = I_{Rf} + (k_o - I_{RF}) Risk Factor

For P-I : RADR = 0.10 + (0.15 – 0.10) 1.80 = 19%

For P-II : RADR = 0.10 + (0.15 – 0.10) 1.00 = 15 %

For P-III : RADR = 0.10 + (0.15 – 0.10) 0.60 = 13 %

(ii) The three projects can now be evaluated at 19%, 15% and 13% discount rate as follows:

Project P-I

Annual Inflows	₹ 6,00,000
PVAF (19 %, 4)	2.639
PV of Inflows (₹ 6,00,000 x 2.639)	₹ 15,83,400

Less: Cost of Investment	₹ 15,00,000
Net Present Value	₹ 83,400

Project P-II

Year	Cash Inflow (₹)	PVF (15%,n)	PV (₹)
1	6,00,000	0.870	5,22,000
2	4,00,000	0.756	3,02,400
3	5,00,000	0.658	3,29,000
4	2,00,000	0.572	1,14,400
Total Present Value			12,67,800
Less: Cost of Investment			11,00,000
Net Present Value			**1,67,800**

Project P-III

Year	Cash Inflow (₹)	PVF (13%,n)	PV (₹)
1	4,00,000	0.885	3,54,000
2	6,00,000	0.783	4,69,800
3	8,00,000	0.693	5,54,400
4	12,00,000	0.613	7,35,600
Total Present Value			21,13,800
Less: Cost of Investment			19,00,000
Net Present Value			**2,13,800**

Project P-III has highest NPV. So, it should be accepted by the firm

Practical Q17

17. A firm has projected the following cash flows from a project under evaluation:

Year	₹ lakhs
0	(70)
1	30
2	40
3	30

The above cash flows have been made at expected prices after recognizing inflation. The firm's cost of capital is 10%. The expected annual rate of inflation is 5%.

Show how the viability of the project is to be evaluated.

Solution — ICAI

17. It is stated that the cash flows have been adjusted for inflation; hence they are "nominal". The cost of capital or discount rate is "real". In order to be compatible, the cash flows should be converted into "real flow". This is done as below:

Year	Nominal cash flows	Adjusted Inflation* factor	Real cash flows	PVF @ 10%	PV of cash flows
0	(70)	–	(70)	1.000	(70)
1	30	0.952	28.56	0.909	25.96
2	40	0.907	36.28	0.826	29.97

3	30	0.864	25.92	0.751	19.47
				Total	75.40
Less: Cash out flow					70.00
NPV (+)					5.40

* $1/1.05$; $1/(1.05)^2$; $1/(1.05)^3$;

Advise: With positive NPV, the project is financially viable.

Alternatively, instead of converting cash flows into real terms, the discount rate can be converted into nominal rate. Result will be the same.

An alternative solution is presented herewith

Alternative solution:

Year	Nominal cash flows	PVF @ 15.50% adjusted by the inflation factor i.e. 5%*	PV of cash flows
1	30	0.866	25.98
2	40	0.749	29.96
3	30	0.649	19.47
		Cash inflow	75.41
		Less: Cash out flow	70.00
		Net present value	5.41

* $\frac{0.909}{1.05}=0.866, \frac{0.826}{1.1025}=0.749, \frac{0.751}{1.1576}=0.649$

Advise: With positive NPV, the project is financially viable.

Practical Q18

18. Shashi Co. Ltd has projected the following cash flows from a project under evaluation:

Year	0	1	2	3
₹ (in lakhs)	(72)	30	40	30

The above cash flows have been made at expected prices after recognizing inflation. The firm's cost of capital is 10%. The expected annual rate of inflation is 5%. Show how the viability of the project is to be evaluated. PVF at 10% for 1-3 years are 0.909, 0.826 and 0.751.

Solution — ICAI

18. Here the given cash flows have to be adjusted for inflation. Alternatively, the discount rate can be converted into nominal rate, as follows:-

Year 1 = $\frac{0.909}{1.05} = 0.866$; Year 2 = $\frac{0.826}{(1.05)^2}$ or $\frac{0.826}{1.1025} = 0.749$

Year 3 = $\frac{0.751}{(1.05)^3} = \frac{0.751}{1.1576} = 0.649$

Year	Nominal Cash Flows (₹ in lakhs)	Adjusted PVF as above	PV of Cash Flows (₹ in lakhs)
1	30	0.866	25.98
2	40	0.749	29.96
3	30	0.649	19.47
	Cash Inflow		75.41
	Less: Cash Outflow		72.00
	Net Present Value		3.41

With positive NPV, the project is financially viable.

Alternative Solution

Assumption: The cost of capital given in the question is "Real'.

Nominal cost of capital = (1.10) (1.05) -1 = 0.155 =15.50%

DCF Analysis of the project

(₹ Lakhs)

	Period	PVF @15.50%	CF	PV
Investment	0	1	-72	-72.00
Operation	1	0.866	30	+25.98
---do---	2	0.750	40	+30.00
---do---	3	0.649	30	+19.47
NPV				+3.45

The proposal may be accepted as the NPV is positive.

Practical Q19

19. KLM Ltd. requires ₹ 15,00,000 for a new project.

Useful life of project is 3 years.

Salvage value - NIL.

Depreciation is ₹ 5,00,000 p.a.

Given below are projected revenues and costs (excluding depreciation) ignoring inflation:

Year →	1	2	3
Revenues in ₹	10,00,000	13,00,000	14,00,000
Costs in ₹	5,00,000	6,00,000	6,50,000

Applicable tax rate is 35%. Assume nominal cost of capital to be 14% (after tax). The inflation rates for revenues and costs are as under:

Year	Revenues %	Costs %
1	9	10
2	8	9
3	6	7

PVF at 14%, for 3 years = 0.877, 0.769 and 0.675

Show amount to the nearest rupee in calculations.

You are required to calculate net present value of the project.

Solution — ICAI

19. (i) Inflation adjusted Revenues

Year	Revenues (₹)	Revenues (Inflation Adjusted) (₹)
1	10,00,000	10,00,000(1.09) = 10,90,000
2	13,00,000	13,00,000(1.09) (1.08) = 15,30,360
3	14,00,000	14,00,000(1.09) (1.08)(1.06) = 17,46,965

(ii) **Inflation adjusted Costs**

Year	Costs (₹)	Costs (Inflation Adjusted) (₹)
1	5,00,000	5,00,000(1.10) = 5,50,000
2	6,00,000	6,00,000(1.10)(1.09) = 7,19,400
3	6,50,000	6,50,000(1.10)(1.09)(1.07) = 8,33,905

(iii) Tax Benefit on Depreciation = ₹ 5,00,000 x 0.35 = ₹ 1,75,000

(iv) **Net Profit after Tax**

Year	Revenues (Inflation Adjusted) (₹) (1)	Costs (Inflation Adjusted) (₹) (2)	Net Profit (₹) (3) =(1) -(2)	Tax (₹) (4) = 35% of (3)	Profit after Tax (₹) (3) - (4)
1	10,90,000	5,50,000	5,40,000	1,89,000	3,51,000
2	15,30,360	7,19,400	8,10,960	2,83,836	5,27,124
3	17,46,965	8,33,905	9,13,060	3,19,571	5,93,489

(v) **Present Value of Cash Inflows**

Year	Net Profit after tax (₹)	Tax Benefit on Depreciation (₹)	Cash Inflow (₹)	PVF@ 14%	PV (₹)
1	3,51,000	1,75,000	5,26,000	0.877	4,61,302
2	5,27,124	1,75,000	7,02,124	0.769	5,39,933
3	5,93,489	1,75,000	7,68,489	0.675	5,18,730
					15,19,965

NPV = ₹ 15,19,965 – ₹ 15,00,000 = ₹ 19,965

Practical Q20

20. A firm has an investment proposal, requiring an outlay of ₹ 80,000. The investment proposal is expected to have two years economic life with no salvage value. In year 1, there is a 0.4 probability that cash inflow after tax will be ₹ 50,000 and 0.6 probability that cash inflow after tax will be ₹ 60,000. The probability assigned to cash inflow after tax for the year 2 is as follows:

The cash inflow year 1	₹ 50,000		₹ 60,000	
The cash inflow year 2	Probability		Probability	
	₹ 24,000	0.2	₹ 40,000	0.4
	₹ 32,000	0.3	₹ 50,000	0.5
	₹ 44,000	0.5	₹ 60,000	0.1

The firm uses a 10% discount rate for this type of investment.

Required:

(i) Construct a decision tree for the proposed investment project and calculate the expected net present value (NPV).

(ii) What net present value will the project yield, if worst outcome is realized? What is the probability of occurrence of this NPV?

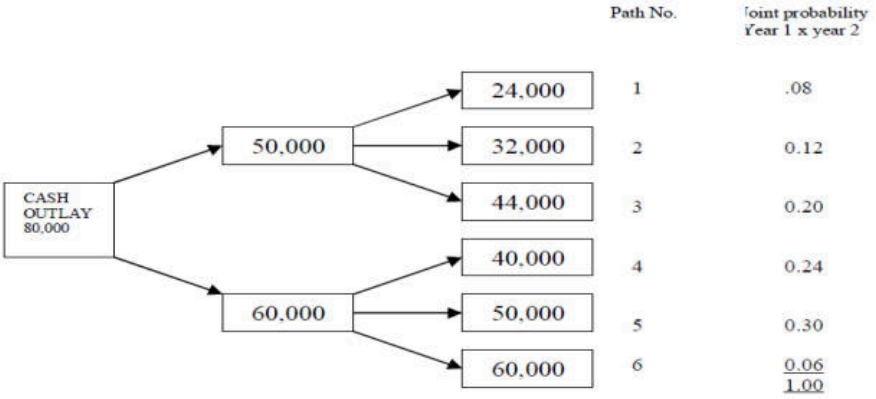
(iii) What will be the best outcome and the probability of that occurrence?

(iv) Will the project be accepted?

(Note: 10% discount factor 1 year 0.909; 2 year 0.826)

Solution — ICAI

20. (i) The decision tree diagram is presented in the chart, identifying various paths and outcomes, and the computation of various paths/outcomes and NPV of each path are presented in the following tables:



The Net Present Value (NPV) of each path at 10% discount rate is given below:

Path	Year 1 Cash Flows (₹)	Year 2 Cash Flows (₹)	Total Cash Inflows (PV) (₹)	Cash Inflows NPV (₹)	(₹)
1	50,000×.909 = 45,450	24,000×.826 = 19,824	65,274	80,000	(–) 14,726
2	45,450	32,000×.826 = 26,432	71,882	80,000	(–) 8,118
3	45,450	44,000×.826 = 36,344	81,794	80,000	1,794
4	60,000×.909 = 54,540	40,000×.826 = 33,040	87,580	80,000	7,580
5	54,540	50,000×.826 = 41,300	95,840	80,000	15,840
6	54,540	60,000×.826 = 49,560	1,04,100	80,000	24,100

Statement showing Expected Net Present Value

₹

z	NPV (₹)	Joint Probability	Expected NPV
1	–14,726	0.08	–1,178.08
2	–8,118	0.12	–974.16
3	1,794	0.20	358.80
4	7,580	0.24	1,819.20
5	15,840	0.30	4,752.00
6	24,100	0.06	1,446.00
			6,223.76

(ii) If the worst outcome is realized the project will yield NPV of – ₹ 14,726. The probability of occurrence of this NPV is 8% and a loss of ₹ 1,178 (path 1).

(iii) The best outcome will be path 6 when the NPV is at ₹ 24,100. The probability of occurrence of this NPV is 6% and a expected profit of ₹ 1,446.

(iv) The project should be accepted because the expected NPV is positive at ₹ 6,223.76 based on joint probability.

Practical Q21

21. Jumble Consultancy Group has determined relative utilities of cash flows of two forthcoming projects of its client company as follows:

Cash Flow in ₹	-15000	-10000	-4000	0	15000	10000	5000	1000
Utilities	-100	-60	-3	0	40	30	20	10

The distribution of cash flows of project A and Project B are as follows:

Project A					
Cash Flow (₹)	-15000	- 10000	15000	10000	5000
Probability	0.10	0.20	0.40	0.20	0.10
Project B					
Cash Flow (₹)	- 10000	-4000	15000	5000	10000
Probability	0.10	0.15	0.40	0.25	0.10

Which project should be selected and why ?

Solution — ICAI

21. Evaluation of project utilizes of Project A and Project B

	Project A		
Cash flow (in ₹)	**Probability**	**Utility**	**Utility value**
-15,000	0.10	-100	-10
-10,000	0.20	-60	-12
15,000	0.40	40	16
10,000	0.20	30	6
5,000	0.10	20	2
			2

Cash flow (in ₹)	Project B		
	Probability	**Utility**	**Utility value**
-10,000	0.10	-60	-6
-4,000	0.15	-3	-0.45
15,000	0.40	40	16
5,000	0.25	20	5
10,000	0.10	30	3
			17.55

Project B should be selected as its expected utility is more.

Practical Q22

22. A & Co. is contemplating whether to replace an existing machine or to spend money on overhauling it. A & Co. currently pays no taxes. The replacement machine costs ₹ 90,000 now and requires maintenance of ₹ 10,000 at the end of every year for eight years. At the end of eight years it would have a salvage value of ₹ 20,000 and would be sold. The existing machine requires increasing amounts of maintenance each year and its salvage value falls each year as follows:

Year	Maintenance (₹)	Salvage (₹)
Present	0	40,000
1	10,000	25,000
2	20,000	15,000
3	30,000	10,000
4	40,000	0

The opportunity cost of capital for A & Co. is 15%.

Required:

When should the company replace the machine?

(Notes: Present value of an annuity of Re. 1 per period for 8 years at interest rate of 15% : 4.4873; present value of Re. 1 to be received after 8 years at interest rate of 15% : 0.3269).

Solution — ICAI

22. **A & Co.**

Equivalent cost of (EAC) of new machine

		₹
(i)	Cost of new machine now	90,000
	Add: PV of annual repairs @ ₹ 10,000 per annum for 8 years (₹ 10,000 × 4.4873)	44,873
		1,34,873
	Less: PV of salvage value at the end of 8 years (₹20,000×0.3269)	6,538
		1,28,335
	Equivalent annual cost (EAC) (₹ 1,28,335/4.4873)	28,600

PV of cost of replacing the old machine in each of 4 years with new machine

Scenario	Year	Cash Flow (₹)	PV @ 15%	PV (₹)
Replace Immediately	0	(28,600)	1.00	(28,600)
		40,000	1.00	40,000
				11,400
Replace in one year	1	(28,600)	0.870	(24,882)
	1	(10,000)	0.870	(8,700)
	1	25,000	0.870	21,750
				(11,832)
Replace in two years	1	(10,000)	0.870	(8,700)
	2	(28,600)	0.756	(21,622)
	2	(20,000)	0.756	(15,120)
	2	15,000	0.756	11,340
				(34,102)
Replace in three years	1	(10,000)	0.870	(8,700)
	2	(20,000)	0.756	(15,120)
	3	(28,600)	0.658	(18,819)
	3	(30,000)	0.658	(19,740)
	3	10,000	0.658	6,580
				(55,799)
Replace in four years	1	(10,000)	0.870	(8,700)
	2	(20,000)	0.756	(15,120)
	3	(30,000)	0.658	(19,740)
	4	(28,600)	0.572	(16,359)
	4	(40,000)	0.572	(22,880)
				(82,799)

Advice: The company should replace the old machine immediately because the PV of cost of replacing the old machine with new machine is least.

Alternatively, optimal replacement period can also be computed using the following table:

Scenario	Year	Cashflow	PV at 15%	PV
Replace immediately	0	(40,000)	1	(40,000)
	1 to 4	28,600	2.855	81,652
				41,652
Replace after 1 year	1	10,000	0.870	8,696
	1	(25,000)	0.870	(21,739)
	2 to 4	28,600	1.985	56,783
				43,739
Replace after 2 years	1	10,000	0.870	8,696
	2	20,000	0.756	15,123
	2	(15,000)	0.756	(11,342)
	3 and 4	28,600	1.229	35,157
				47,633
Replace after 3 years	1	10,000	0.870	8,696
	2	20,000	0.756	15,123
	3	30,000	0.658	19,725
	3	(10,000)	0.658	(6,575)
	4	28,600	0.572	16,352
				53,321
Replace after 4 years	1	10,000	0.870	8,696
	2	20,000	0.756	15,123
	3	30,000	0.658	19,725
	4	40,000	0.572	22,870
				66,414

Practical Q23

23. A company has an old machine having book value zero – which can be sold for ₹ 50,000. The company is thinking to choose one from following two alternatives:

(i) To incur additional cost of ₹ 10,00,000 to upgrade the old existing machine.

(ii) To replace old machine with a new machine costing ₹ 20,00,000 plus installation cost ₹ 50,000.

Both above proposals envisage useful life to be five years with salvage value to be nil.

The expected after tax profits for the above three alternatives are as under :

Year	Old existing Machine (₹)	Upgraded Machine (₹)	New Machine (₹)
1	5,00,000	5,50,000	6,00,000
2	5,40,000	5,90,000	6,40,000
3	5,80,000	6,10,000	6,90,000
4	6,20,000	6,50,000	7,40,000
5	6,60,000	7,00,000	8,00,000

The tax rate is 40 per cent.

The company follows straight line method of depreciation. Assume cost of capital to be 15 per cent.

P.V.F. of 15%, 5 = 0.870, 0.756, 0.658, 0.572 and 0.497. You are required to advise the company as to which alternative is to be adopted.

Solution — ICAI

23.

(A)	Cash Outflow		₹
	(i)	In case machine is upgraded:	
		Upgradation Cost	10,00,000
	(ii)	In case new machine installed:	
		Cost	20,00,000
		Add: Installation cost	50,000
		Total Cost	20,50,000
		Less: Disposal of old machine	
		₹ 50,000 – 40% tax	30,000
		Total Cash Outflow	20,20,000

Working Note:

(i) Depreciation – in case machine is upgraded

₹ 10,00,000 ÷ 5 = ₹ 2,00,000

(ii) Depreciation – in case new machine is installed

₹ 20,50,000 ÷ 5 = ₹ 4,10,000

(iii) Old existing machine – Book Value is zero. So, no depreciation.

(B) Cash Inflows after Taxes (CFAT)

Year	Old Existing Machine	Upgraded Machine			
	(i) EAT/CFAT ₹	(ii) EAT ₹	(iii) DEP ₹	(iv) CFAT ₹	= (iv)-(i) Incremental CFAT ₹
1	5,00,000	5,50,000	2,00,000	7,50,000	2,50,000
2	5,40,000	5,90,000	2,00,000	7,90,000	2,50,000
3	5,80,000	6,10,000	2,00,000	8,10,000	2,30,000
4	6,20,000	6,50,000	2,00,000	8,50,000	2,30,000
5	6,60,000	7,00,000	2,00,000	9,00,000	2,40,000

Cash Inflow after Taxes (CFAT)

Year		New Machine		
	(vi) EAT ₹	(vii) DEP ₹	(viii) CFAT ₹	(ix) = (viii) – (i) Incremental CFAT (₹)
1	6,00,000	4,10,000	10,10,000	5,10,000
2	6,40,000	4,10,000	10,50,000	5,10,000
3	6,90,000	4,10,000	11,00,000	5,20,000
4	7,40,000	4,10,000	11,50,000	5,30,000
5	8,00,000	4,10,000	12,10,000	5,50,000

P.V. AT 15% - 5 Years – on Incremental CFAT

Year	Upgraded Machine			New Machine		
	Incremental CFAT ₹	PVF	Total P.V. ₹	Incremental CFAT	PVF	Total PV ₹
1	2,50,000	0.870	2,17,500	5,10,000	0.870	4,43,700
2	2,50,000	0.756	1,89,000	5,10,000	0.756	3,85,560
3	2,30,000	0.658	1,51,340	5,20,000	0.658	3,42,160
4	2,30,000	0.572	1,31,560	5,30,000	0.572	3,03,160
5.	2,40,000	0.497	1,19,280	5,50,000	0.497	2,73,350
Total P.V. of CFAT			8,08,680			17,47,930
Less: Cash Outflows			10,00,000			20,20,000*
N.P.V. =			-1,91,320			- 2,72,070

*Acquisition Cost (including installation cost)	₹ 20,50,000
Less: Salvage Value of existing machine net of Tax	₹ 30,000
	₹ 20,20,000

As the NPV in both the new (alternative) proposals is negative, the company should continue with the existing old Machine.

Practical Q24

24. Company X is forced to choose between two machines A and B. The two machines are designed differently but have identical capacity and do exactly the same job. Machine A costs ₹ 1,50,000 and will last for 3 years. It costs ₹ 40,000 per year to run. Machine B is an 'economy' model costing only ₹ 1,00,000, but will last only for 2 years, and costs ₹ 60,000 per year to run. These are real cash flows. The costs are forecasted in rupees of constant purchasing power. Ignore tax. Opportunity cost of capital is 10 per cent. Which machine company X should buy?

Solution — ICAI

24. Statement showing the evaluation of two machines

Machines	A	B
Purchase cost (₹): (i)	1,50,000	1,00,000
Life of machines (years)	3	2
Running cost of machine per year (₹): (ii)	40,000	60,000
Cumulative present value factor for 1-3 years @ 10% (iii)	2.486	–
Cumulative present value factor for 1-2 years @ 10% (iv)	–	1.735
Present value of running cost of machines (₹): (v)	99,440 [(ii) × (iii)]	1,04,100 [(ii) × (iv)]
Cash outflow of machines (₹): (vi) = (i) + (v)	2,49,440	2,04,100
Equivalent present value of annual cash outflow	1,00,338 [(vi) ÷ (iii)]	1,17,637 [(vi) ÷ (iv)]

Decision: Company X should buy machine A since its equivalent cash outflow is less than machine B.

Practical Q25

25. Company Y is operating an elderly machine that is expected to produce a net cash inflow of ₹ 40,000 in the coming year and ₹ 40,000 next year. Current salvage value is ₹ 80,000 and next year's value is ₹ 70,000. The machine can be replaced now with a new machine, which costs ₹ 1,50,000, but is much more efficient and will provide a cash inflow of ₹ 80,000 a year for 3 years. Company Y wants to know whether it should replace the equipment now or wait a year with the clear understanding that the new machine is the best of the available alternatives and that it in turn be replaced at the optimal point. Ignore tax. Take opportunity cost of capital as 10 per cent. Advise with reasons.

Solution — ICAI

25. Statement showing present value of cash inflow of new machine when it replaces elderly machine now

NPV of New Machine

PV of Cash Inflow (80000 x 2.486)	₹ 1,98,880
Less: Purchase Cost of New Machine	₹ 1,50,000
	₹ 48,880

Since NPV of New Machine is positive, it should be purchased.

Timing Decision

Replace Now

Current Realizable Value	₹ 80,000
NPV of New Machine	₹ 48,880
Total NPV	₹ 1,28,880

Replace after 1 Year

Cash Inflow for Year 1	₹ 40000
Realisable Value of Old Machine	₹ 70000

NPV of New Machine	₹ 48,880
Total NPV after 1 Year	₹ 1,58,880
PV of Total NPV (158880/1.1)	₹ 1,44,436

Advise: Since Total NPV is higher in case of Replacement after one year Machine should be replaced after 1 year.

Practical Q26

26. A machine used on a production line must be replaced at least every four years. Costs incurred to run the machine according to its age are:

Age of the Machine (years)					
	0	1	2	3	4
Purchase price (in ₹)	60,000				
Maintenance (in ₹)		16,000	18,000	20,000	20,000
Repair (in ₹)		0	4,000	8,000	16,000
Scrap Value (in ₹)		32,000	24,000	16,000	8,000

Future replacement will be with identical machine with same cost. Revenue is unaffected by the age of the machine. Ignoring inflation and tax, determine the optimum replacement cycle. PV factors of the cost of capital of 15% for the respective four years are 0.8696, 0.7561, 0.6575 and 0.5718.

Solution — ICAI

26. Working Notes

First of all, we shall calculate cash flows for each replacement cycle as follows:

One Year Replacement Cycle ₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	32,000	16,000

Two Years Replacement Cycle ₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	-	(16,000)
2	-	(22,000)	24,000	2,000

Three Years Replacement Cycle ₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	-	(16,000)
2	-	(22,000)	-	(22,000)
3	-	(28,000)	16,000	(12,000)

Four Years Replacement Cycle ₹

Year	Replacement Cost	Maintenance & Repair	Residual Value	Net cash Flow
0	(60,000)	-	-	(60,000)
1	-	(16,000)	-	(16,000)
2	-	(22,000)	-	(22,000)
3	-	(28,000)	-	(28,000)
4	-	(36,000)	8,000	(28,000)

Now we shall calculate NPV for each replacement cycles

		1 Year		2 Years		3 Years		4 Years	
Year	PVF@ 15%	Cash Flows	PV	Cash Flows	PV	Cash Flows	PV	Cash Flows	PV
0	1	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000	-60,000
1	0.8696	16,000	13,914	-16,000	-13,914	-16,000	-13,914	-16,000	-13,914
2	0.7561	-	-	2,000	1,512	-22,000	-16,634	-22,000	-16,634
3	0.6575	-	-	-	0	-12,000	-7,890	-28,000	-18,410
4	0.5718	-	-	-	0		0	-28,000	-16,010
			-46,086		-72,402		-98,438		-1,24,968

Replacement Cycles		EAC (₹)
1 Year	46,086 / 0.8696	52,997
2 Years	72,402 / 1.6257	44,536
3 Years	98,438 / 2.2832	43,114
4 Years	1,24,968 / 2.855	43,772

Since EAC is least in case of replacement cycle of 3 years hence machine should be replaced after every three years.

Practical Q27

27. Trouble Free Solutions (TFS) is an authorized service center of a reputed domestic air conditioner manufacturing company. All complaints/service related matters of Air conditioner are attended by this service center. The service center employs a large number of mechanics, each of whom is provided with a motor bike to attend the complaints. Each mechanic travels approximately 40000 kms per annuam. TFS decides to continue its present policy of always buying a new bike for its mechanics but wonders whether the present policy of replacing the bike every three year is optimal or not. It is of believe that as new models are entering into market on yearly basis, it wishes to consider whether a replacement of either one year or two years would be better option than present three year period. The fleet of bike is due for replacement shortly in near future.

The purchase price of latest model bike is ₹ 55,000. Resale value of used bike at current prices in market is as follows:

Period	₹
1 Year old	35,000
2 Year old	21,000
3 Year old	9,000

Running and Maintenance expenses (excluding depreciation) are as follows:

Year	Road Taxes Insurance etc. (₹)	Petrol Repair Maintenance etc. (₹)
1	3,000	30,000
2	3,000	35,000
3	3,000	43,000

Using opportunity cost of capital as 10% you are required to determine optimal replacement period of bike.

Solution — ICAI

27. In this question the effect of increasing running cost and decreasing resale value have to be weighted upto against the purchase cost of bike. For this purpose, we shall compute Equivalent Annual Cost (EAC) of replacement in different years shall be computed and compared.

Year	Road Taxes (₹)	Petrol etc. (₹)	Total (₹)	PVF @10%	PV (₹)	Cumulative PV (₹)	PV of Resale Price (₹)	Net Outflow (₹)
1	3,000	30,000	33,000	0.909	29,997	29,997	31,815	(1,818)
2	3,000	35,000	38,000	0.826	31,388	61,385	17,346	44,039
3	3,000	43,000	46,000	0.751	34,546	95,931	6,759	89,172

Computation of EACs

Year*	Purchase Price of Bike (₹)	Net Outflow (₹)	Total Outflow (₹)	PVAF @ 10%	EAC♣ (₹)
1	55,000	(1,818)	53,182	0.909	58,506
2	55,000	44,039	99,039	1.735	57,083
3	55,000	89,172	1,44,172	2.486	57,993

Thus, from above table it is clear that EAC is least in case of 2 years, hence bike should be replaced every two years.

* Assume these periods are the periods from which bike shall be kept in use.

♣ EAC is used to bring Cash Flows occurring for different periods at one point of Time.

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